

Causal Inference

Fortunate is one who is able to know the causes of things.

—VIRGIL

What is causal inference? Recall that a feature of a probability distribution is identified if it can be inferred given full knowledge of the distribution of observables. Causal identification is just identification of any feature of a distribution that can be interpreted as a causal effect. Causal inference, then, is the process of estimating these quantities from data.¹

Major empirical debates in the social and health sciences often revolve around causal questions. Examples abound: Does smoking cause lung cancer, and if so, how much might we expect smoking cessation to increase longevity? Do cash transfers to the poor reduce labor supply and, if so, by how much? All else equal, how much does a job applicant's reported racial or gender identity affect the probability of receiving an interview? These questions have sparked considerable controversy in their respective fields. One reason that these and similar questions do, in fact, yield debates is that even causal questions of these types cannot be answered without making assumptions external to the statistical model; controversies cannot be resolved via statistical tools alone.

In this chapter, we discuss the kinds of assumptions necessary to answer causal questions through statistical models. We show that inferences about causal questions—causal inferences—rest on a fundamental identification challenge: causal questions can be viewed as an attempt to make a comparison between two or more hypothetical states of the world, yet we can only ever observe one. In order to proceed, we must be willing to make assumptions about quantities that are not observed by the researcher. This task

¹ In this book, we have generally used the term “inference” to refer specifically to the quantification of uncertainty. However, the term “causal inference” is commonly used in this broader sense, encompassing both estimation and inference, and we will adhere to this established convention.

can be shown to be separable from the statistical assumptions necessary to characterize observed data.

Readers will note that the structure and language of this chapter largely mirrors (often near-identically) that of Chapter 6. This is no accident, and as we proceed, we will show how the identification results and estimation procedures that we derived for missing data are easily translated to address questions of cause and effect. Again, to highlight the parallels across chapters, section numbers in brackets will reference the analogous section in Chapter 6.

7.1 IDENTIFICATION WITH POTENTIAL OUTCOMES [6.1]

We employ a model of *potential outcomes* (Splawa-Neyman, Dabrowska, and Speed 1923) known as the *Neyman–Rubin Causal Model (NRCM)*.² The NRCM is a powerful and flexible framework for describing claims of cause and effect. There are other formalizations of causality, but the NRCM is the most commonly used framework in political science, biostatistics, and many branches of applied econometrics.

Suppose that we want to know the effect of a medication on a patient’s self-reported level of pain. The problem is that we cannot observe a person’s counterfactual outcome. If the patient takes the medication, we cannot know what they would have reported if they had not. If the patient does not take the medication, we cannot know what they would have reported if they had. This is known as the *Fundamental Problem of Causal Inference* (Holland 1986).

7.1.1 Framework

Suppose that we have a binary treatment D_i , so that $D_i = 1$ if unit i receives the treatment and $D_i = 0$ otherwise. Then the NRCM can be formalized as follows:

Definition 7.1.1. *Potential Outcomes Model*

For an outcome variable Y_i and binary treatment variable D_i , the *potential outcomes model* is

$$Y_i = \begin{cases} Y_i(0) & : D_i = 0 \\ Y_i(1) & : D_i = 1 \end{cases}$$

$$= Y_i(1) \cdot D_i + Y_i(0) \cdot (1 - D_i),$$

where $Y_i(0)$ denotes the potential outcome under control for unit i , and $Y_i(1)$ denotes the potential outcome under treatment for unit i .

² Also sometimes referred to as the Rubin Causal Model.

Throughout this section, we will assume that the potential outcomes model holds. Note that the distribution of (Y_i, D_i) is observable, but the distribution of $(Y_i(0), Y_i(1))$ is not directly observable. Thus, this is an *identification* problem (see Definition 6.0.1). Potential outcomes $Y_i(d)$ (for $d \in \{0, 1\}$) can be binary, discrete, continuous, etc., as long as $\text{Supp}[Y_i(d)] \subseteq \mathbb{R}$. The population values of $(Y_i(0), Y_i(1), D_i)$ can have any joint distribution (subject to perhaps mild regularity conditions for identification and estimation).

Analogously to the *stable outcome model* for missing data (Definition 6.1.1), the above equation embeds what is known as the *stable unit treatment value assumption*. Potential outcomes are assumed to be stable. No matter what, for every unit i , when $D_i = 1$, we observe $Y_i(1)$, and when $D_i = 0$, we observe $Y_i(0)$. This implies that (1) there are no unobserved multiple versions of the treatment or control—every unit i has a single outcome value $Y_i(1)$ that is observed when $D_i = 1$, and a single outcome value $Y_i(0)$ that is observed when $D_i = 0$; and (2) there is no interference between units—unit i 's potential outcomes are not affected by whether or not any other unit j receives the treatment.³

The potential outcomes model allows us to define some natural quantities as targets of inquiry. One foundational quantity is the (unobservable) treatment effect associated with each unit.

Definition 7.1.2. Unit-Level Treatment Effect

Under the potential outcomes model, the *causal effect of treatment* for unit i is

$$\tau_i = Y_i(1) - Y_i(0).$$

Note that τ_i is a random variable that reflects the distribution of treatment effects in the population. Again, the Fundamental Problem of Causal Inference is that we can never observe τ_i directly, since we can never observe both $Y_i(0)$ and $Y_i(1)$ for a given unit i .⁴

Our primary inferential target will be the expected value of this distribution of treatment effects. This expected value is known as the *average treatment effect (ATE)*, also known as the average causal effect (ACE).

³ Violations of the stable unit treatment value assumption do not rule out interesting work. There are generalizations of the NRCM that imply that all our calculations will be meaningful even when the stable unit treatment value assumption does not hold, though the interpretation is somewhat more complex. See, e.g., Hudgens and Halloran (2008) for a discussion of how typical methods estimate regime-specific causal effects when there is interference between units.

⁴ It might seem like we could solve this problem by simply observing the potential outcomes sequentially: observe $Y_i(0)$, and then administer the treatment and observe $Y_i(1)$. However, this requires the assumption that unit i does not change over time or as a result of the act of measurement. This assumption may be justifiable in some laboratory experiments in the natural sciences where conditions can be precisely controlled, but it is rarely a safe assumption to make in the social or health sciences. We also might be interested in outcomes that can only be measured once, such as which candidate a voter votes for in a particular election.

Definition 7.1.3. *Average Treatment Effect*

Under the potential outcomes model, the *average treatment effect (ATE)* is $E[\tau_i]$.

Some texts refer to this quantity as the population average treatment effect (PATE), since we are assuming independent and identically distributed (i.i.d.) sampling from some population.⁵ Before continuing, we note an important direct result about the ATE that will be central to our discussion in Section 7.1.4.

Theorem 7.1.4. *ATE Decomposition*

Under the potential outcomes model, the ATE can be written as

$$E[\tau_i] = E[Y_i(1) - Y_i(0)] = E[Y_i(1)] - E[Y_i(0)].$$

The proof follows immediately from linearity of expectations. This means that the ATE can be written simply as the difference between the average potential outcome in treatment and the average potential outcome in control.

There are other causal targets that researchers have in mind, including the *average treatment effect among the treated (ATT)* ($E[\tau_i | D_i = 1]$) or the *average treatment effect among the control subjects (ATC)* ($E[\tau_i | D_i = 0]$). These targets are often of policy interest—the ATT, for instance, corresponds to the average effect of treatment among the (potentially non-representative) subpopulation of subjects who actually receive treatment, thus speaking to the real-world effect of an existing treatment. In addition, the ATT and ATC can each be point identified under weaker assumptions than the ATE. We focus here on the ATE, though our calculations would generalize straightforwardly via conditioning on D_i . We will return to briefly discuss the ATT as an inferential target, as well as others, in Section 7.3.

7.1.2 Ties to Missing Data

We do *not* directly observe i.i.d. draws of $Y_i(0)$ or $Y_i(1)$, and we *never* observe τ_i for any unit i . (This is the Fundamental Problem of Causal Inference.) We only observe i.i.d. draws of the random vector (Y_i, D_i) . So the question is: given full knowledge of the distribution of (Y_i, D_i) , what assumptions yield what information about the distributions of $Y_i(0)$, $Y_i(1)$, and τ_i ?

⁵ There exists an alternative formulation of causal inference under the NRCM that does *not* require the assumption of i.i.d. sampling, known as *design-based inference*. In this book, we focus on causal inference in the i.i.d. sampling context, as it considerably eases exposition.

Causal inference under the NRCM is a missing data problem.⁶ We only learn about $Y_i(1)$ for treated units and we only learn about $Y_i(0)$ for control units.

Example 7.1.5. *Causal Inference as a Missing Data Problem*

Suppose we had binary outcomes: $\text{Supp}[Y_i(0)] \subseteq \{0, 1\}$ and $\text{Supp}[Y_i(1)] \subseteq \{0, 1\}$, and therefore $\text{Supp}[Y_i] \subseteq \{0, 1\}$. We might see the following data:

Unit	Y_i	D_i
1	1	1
2	1	0
3	1	1
4	0	1
5	1	1
6	0	0

Given the potential outcomes model, this implies

Unit	$Y_i(0)$	$Y_i(1)$	D_i
1	?	1	1
2	1	?	0
3	?	1	1
4	?	0	1
5	?	1	1
6	0	?	0

We want to learn about the distributions of $Y_i(0)$ and $Y_i(1)$. But we have a missing data problem. We can identify the joint distribution of (Y_i, D_i) values—with large n , we could learn about the distribution of (Y_i, D_i) to arbitrary precision. But even with large n , we need additional assumptions to learn about the distributions of $Y_i(0)$ and $Y_i(1)$. It is even harder to learn about τ_i .

Recognizing that the Fundamental Problem of Causal Inference is essentially a missing data problem helps us to understand the types of assumptions that we need in order to estimate causal effects. Causal inference under the NRCM is about coming up with ways of “solving” this missing data problem with further assumptions. The approaches that we discuss in this chapter are completely analogous to those we presented for missing data in Chapter 6. Δ

⁶ This insight is attributed to Rubin (1974), which is a major reason why we have the “R” in NRCM. The converse, however, is also true. Missing data problems can also be viewed as a causal inference problem, an insight attributable to Robins (1986). To see this, we need only recognize that we are asking a question about a potential outcome that we did not observe. Missing data questions, wedded to an assumption of stable outcomes in Section 6.1 (*Identification with Missing Data*), are questions about counterfactual worlds.

7.1.3 Bounds [6.1.1]

We can apply bounds to this problem. We only assume that potential outcomes are bounded: $\forall d \in \{0, 1\}, \text{Supp}[Y_i(d)] \subseteq [a, b]$, for some $a \leq b$. How can we estimate bounds for $E[\tau_i]$? We can apply the same principles that we applied in the case of missing data, as demonstrated in the following example.

Example 7.1.6. *Bounding Treatment Effects*

In Example 7.1.5, outcomes are binary, so $\forall d \in \{0, 1\}, \text{Supp}[Y_i(d)] \subseteq \{0, 1\} \subseteq [0, 1]$. For the treated units (those with $D_i = 1$), we observe $Y_i(1)$, but we only know that $0 \leq Y_i(0) \leq 1$. Conversely, for the control units (those with $D_i = 0$), we observe $Y_i(0)$, but we only know that $0 \leq Y_i(1) \leq 1$. We know that $E[\tau_i] = E[Y_i(1)] - E[Y_i(0)]$, so an upper bound is obtained by estimating an upper bound for $E[Y_i(1)]$ and a lower bound for $E[Y_i(0)]$. These are computed in the standard way, by plugging in one for the missing values of $Y_i(1)$ and zero for the missing values of $Y_i(0)$:

Unit	$Y_i(0)$	$Y_i(1)$	D_i
1	0	1	1
2	1	1	0
3	0	1	1
4	0	0	1
5	0	1	1
6	0	1	0

Then, using the sample mean as a plug-in estimator for the expected value, our estimate of an upper bound for $E[Y_i(1)]$ is $\frac{5}{6}$, and our estimate of a lower bound for $E[Y_i(0)]$ is $\frac{1}{6}$. Thus, our estimate of an upper bound for $E[\tau_i]$ is $\frac{5}{6} - \frac{1}{6} = \frac{4}{6}$.

Likewise, to estimate a lower bound for $E[\tau_i]$, we estimate a lower bound for $E[Y_i(1)]$ and an upper bound for $E[Y_i(0)]$:

Unit	$Y_i(0)$	$Y_i(1)$	D_i
1	1	1	1
2	1	0	0
3	1	1	1
4	1	0	1
5	1	1	1
6	0	0	0

Our estimated lower bound for $E[Y_i(1)]$ is $\frac{3}{6}$, and our estimated upper bound for $E[Y_i(0)]$ is $\frac{5}{6}$. Thus, our estimated lower bound for $E[\tau_i]$ is $\frac{3}{6} - \frac{5}{6} = -\frac{2}{6}$. Δ

More generally, we can write down exact sharp bounds for $E[\tau_i]$ in terms of the joint distribution of (Y_i, D_i) , which can, in theory, be estimated to arbitrary precision with large n .

Theorem 7.1.7. *Sharp Bounds for the ATE*

Let $Y_i(0)$, $Y_i(1)$, and D_i be random variables such that, $\forall d \in \{0, 1\}$, $\text{Supp}[Y_i(d)] \subseteq [a, b]$, and $\text{Supp}[D_i] = \{0, 1\}$. Let $Y_i = Y_i(1) \cdot D_i + Y_i(0) \cdot (1 - D_i)$ and $\tau_i = Y_i(1) - Y_i(0)$. Then

$$\begin{aligned} E[\tau_i] \in & \left[E[Y_i | D_i = 1] \Pr[D_i = 1] + a \Pr[D_i = 0] \right. \\ & \left. - (E[Y_i | D_i = 0] \Pr[D_i = 0] + b \Pr[D_i = 1]), \right. \\ & E[Y_i | D_i = 1] \Pr[D_i = 1] + b \Pr[D_i = 0] \\ & \left. - (E[Y_i | D_i = 0] \Pr[D_i = 0] + a \Pr[D_i = 1]) \right]. \end{aligned}$$

Proof: Our proof follows from Theorem 6.1.3 (*Sharp Bounds for the Expected Value*).

By the Law of Iterated Expectations,

$$E[Y_i(1)] = E[Y_i(1) | D_i = 1] \Pr[D_i = 1] + E[Y_i(1) | D_i = 0] \Pr[D_i = 0].$$

Then, since $Y_i = Y_i(1) \cdot D_i + Y_i(0) \cdot (1 - D_i)$,

$$\begin{aligned} E[Y_i | D_i = 1] &= E[Y_i(1) \cdot D_i + Y_i(0) \cdot (1 - D_i) | D_i = 1] \\ &= E[Y_i(1) \cdot 1 + Y_i(0) \cdot 0 | D_i = 1] \\ &= E[Y_i(1) | D_i = 1]. \end{aligned}$$

Thus,

$$E[Y_i(1)] = E[Y_i | D_i = 1] \Pr[D_i = 1] + E[Y_i(1) | D_i = 0] \Pr[D_i = 0].$$

Since $\text{Supp}[Y_i(1)] \subseteq [a, b]$, $E[Y_i(1) | D_i = 0] \geq a$, and therefore

$$\begin{aligned} E[Y_i(1)] &= E[Y_i | D_i = 1] \Pr[D_i = 1] + E[Y_i(1) | D_i = 0] \Pr[D_i = 0] \\ &\geq E[Y_i | D_i = 1] \Pr[D_i = 1] + a \Pr[D_i = 0]. \end{aligned}$$

Likewise, since $\text{Supp}[Y_i(1)] \subseteq [a, b]$, $E[Y_i(1) | D_i = 0] \leq b$, and therefore

$$\begin{aligned} E[Y_i(1)] &= E[Y_i | D_i = 1] \Pr[D_i = 1] + E[Y_i(1) | D_i = 0] \Pr[D_i = 0] \\ &\leq E[Y_i | D_i = 1] \Pr[D_i = 1] + b \Pr[D_i = 0]. \end{aligned}$$

Thus,

$$\begin{aligned} E[Y_i(1)] \in & [E[Y_i | D_i = 1] \Pr[D_i = 1] + a \Pr[D_i = 0], \\ & E[Y_i | D_i = 1] \Pr[D_i = 1] + b \Pr[D_i = 0]]. \end{aligned}$$

Following parallel steps to bound $E[Y_i(0)]$, we get

$$E[Y_i(0)] \in [E[Y_i|D_i = 0] \Pr[D_i = 0] + a \Pr[D_i = 1], \\ E[Y_i|D_i = 0] \Pr[D_i = 0] + b \Pr[D_i = 1]].$$

Again, by the Law of Iterated Expectations, and substituting in the results from above,

$$E[\tau_i] = E[Y_i(1) - Y_i(0)] \\ = E[Y_i(1)] - E[Y_i(0)] \\ = E[Y_i|D_i = 1] \Pr[D_i = 1] + E[Y_i(1) | D_i = 0] \Pr[D_i = 0] \\ - \left(E[Y_i|D_i = 0] \Pr[D_i = 0] + E[Y_i(0) | D_i = 1] \Pr[D_i = 1] \right).$$

Combining the bounds we found above, since $\text{Supp}[Y_i(d)] \subseteq [a, b]$, $E[Y_i(1)|D_i = 0] \geq a$ and $E[Y_i(0)|D_i = 1] \leq b$. Therefore, we can put a lower bound on $E[\tau_i]$:

$$E[\tau_i] = E[Y_i|D_i = 1] \Pr[D_i = 1] + E[Y_i(1) | D_i = 0] \Pr[D_i = 0] \\ - \left(E[Y_i|D_i = 0] \Pr[D_i = 0] + E[Y_i(0) | D_i = 1] \Pr[D_i = 1] \right) \\ \geq E[Y_i|D_i = 1] \Pr[D_i = 1] + a \Pr[D_i = 0] \\ - \left(E[Y_i|D_i = 0] \Pr[D_i = 0] + b \Pr[D_i = 1] \right).$$

Likewise, $E[Y_i(1)|D_i = 0] \leq b$, and $E[Y_i(0)|D_i = 1] \geq a$, and we can put an upper bound on $E[\tau_i]$:

$$E[\tau_i] = E[Y_i|D_i = 1] \Pr[D_i = 1] + E[Y_i(1) | D_i = 0] \Pr[D_i = 0] \\ - \left(E[Y_i|D_i = 0] \Pr[D_i = 0] + E[Y_i(0) | D_i = 1] \Pr[D_i = 1] \right) \\ \leq E[Y_i|D_i = 1] \Pr[D_i = 1] + b \Pr[D_i = 0] \\ - \left(E[Y_i|D_i = 0] \Pr[D_i = 0] + a \Pr[D_i = 1] \right).$$

Thus,

$$E[\tau_i] \in \left[E[Y_i|D_i = 1] \Pr[D_i = 1] + a \Pr[D_i = 0] \right. \\ \left. - \left(E[Y_i|D_i = 0] \Pr[D_i = 0] + b \Pr[D_i = 1] \right), \right. \\ \left. E[Y_i|D_i = 1] \Pr[D_i = 1] + b \Pr[D_i = 0] \right. \\ \left. - \left(E[Y_i|D_i = 0] \Pr[D_i = 0] + a \Pr[D_i = 1] \right) \right]. \quad \square$$

One interesting feature of these bounds is that their width is always $(b - a)$, and does not depend on any other feature of the data. Our discussion

following Theorem 6.1.3 noted that the bounds in the missing data case were proportional to the amount of missing data. In the causal inference setting, the amount of “missing data” in terms of potential outcomes is always the same, as we always observe only one per subject.

Estimation of these bounds again follows from the plug-in principle (Section 3.3) and is again unbiased and consistent.

Theorem 7.1.8. *Estimating Sharp Bounds for the ATE*

Let $Y_i(0)$, $Y_i(1)$, and D_i be random variables such that, $\forall d \in \{0, 1\}$, $\text{Supp}[Y_i(d)] \subseteq [a, b]$, and $\text{Supp}[D_i] = \{0, 1\}$. Let $Y_i = Y_i(1) \cdot D_i + Y_i(0) \cdot (1 - D_i)$ and $\tau_i = Y_i(1) - Y_i(0)$. Then, given n i.i.d. observations of (Y_i, D_i) , the plug-in estimator

$$\frac{1}{n} \sum_{i=1}^n (Y_i D_i + a(1 - D_i) - Y_i(1 - D_i) - bD_i)$$

is unbiased and consistent for the lower bound for $E[\tau_i]$. Likewise, the plug-in estimator

$$\frac{1}{n} \sum_{i=1}^n (Y_i D_i + b(1 - D_i) - Y_i(1 - D_i) - aD_i)$$

is unbiased and consistent for the upper bound for $E[\tau_i]$.

Proof: Our proof follows directly from Theorem 6.1.4 (*Estimating Sharp Bounds for the Expected Value*). Let $\tau_i^L = Y_i D_i + a(1 - D_i) - Y_i(1 - D_i) - bD_i$, so that

$$\overline{\tau_i^L} = \frac{1}{n} \sum_{i=1}^n (Y_i D_i + a(1 - D_i) - Y_i(1 - D_i) - bD_i).$$

By Theorem 3.2.3 (*The Expected Value of the Sample Mean is the Population Mean*) and the Weak Law of Large Numbers (WLLN), $\overline{\tau_i^L}$ is unbiased and consistent for $E[\tau_i^L]$, and by the Law of Iterated Expectations,

$$\begin{aligned} E[\tau_i^L] &= E[\tau_i^L \mid D_i = 1] \Pr[D_i = 1] + E[\tau_i^L \mid D_i = 0] \Pr[D_i = 0] \\ &= E[Y_i D_i + a(1 - D_i) - Y_i(1 - D_i) - bD_i \mid D_i = 1] \Pr[D_i = 1] \\ &\quad + E[Y_i D_i + a(1 - D_i) - Y_i(1 - D_i) - bD_i \mid D_i = 0] \Pr[D_i = 0] \\ &= E[Y_i \cdot 1 + a \cdot 0 - Y_i \cdot 0 - b \cdot 1 \mid D_i = 1] \Pr[D_i = 1] \\ &\quad + E[Y_i \cdot 0 + a \cdot 1 - Y_i \cdot 1 - b \cdot 0 \mid D_i = 0] \Pr[D_i = 0] \end{aligned}$$

$$\begin{aligned}
&= E[Y_i - b | D_i = 1] \Pr[D_i = 1] \\
&\quad + E[a - Y_i | D_i = 0] \Pr[D_i = 0] \\
&= E[Y_i | D_i = 1] \Pr[D_i = 1] - b \Pr[D_i = 1] + a \Pr[D_i = 0] \\
&\quad - E[Y_i | D_i = 0] \Pr[D_i = 0] \\
&= E[Y_i | D_i = 1] \Pr[D_i = 1] + a \Pr[D_i = 0] \\
&\quad - (E[Y_i | D_i = 0] \Pr[D_i = 0] + b \Pr[D_i = 1]).
\end{aligned}$$

Thus, $\overline{\tau}_i^L$ is unbiased and consistent for the lower bound of $E[\tau_i]$, $E[Y_i | D_i = 1] \Pr[D_i = 1] + a \Pr[D_i = 0] - (E[Y_i | D_i = 0] \Pr[D_i = 0] + b \Pr[D_i = 1])$.

Similarly, let $\tau_i^U = Y_i D_i + b(1 - D_i) - Y_i(1 - D_i) - aD_i$, so that

$$\overline{\tau}_i^U = \frac{1}{n} \sum_{i=1}^n (Y_i D_i + b(1 - D_i) - Y_i(1 - D_i) - aD_i).$$

Then, by the same logic, it follows that $\overline{\tau}_i^U$ is unbiased and consistent for $E[Y_i | D_i = 1] \Pr[D_i = 1] + b \Pr[D_i = 0] - (E[Y_i | D_i = 0] \Pr[D_i = 0] + a \Pr[D_i = 1])$. $\square$

Under the usual plug-in regularity conditions, asymptotically valid confidence intervals for each bound can be computed using the bootstrap.

Without adding assumptions, these bounds are tight, and therefore $E[\tau_i]$ is only partially identified. We now ask: what type of assumptions do we need to achieve point identification of $E[\tau_i]$? The answer is completely analogous to the case of missing data.

7.1.4 Random Assignment [6.1.2]

It is generally thought that one of the strongest nonparametric assumptions we could impose is *random assignment* of the treatment. This assumption is analogous to missing completely at random (MCAR).

Definition 7.1.9. *Random Assignment*

Let $Y_i(0)$, $Y_i(1)$, and D_i be random variables with $\text{Supp}[D_i] = \{0, 1\}$. Let $Y_i = Y_i(1) \cdot D_i + Y_i(0) \cdot (1 - D_i)$. Then D_i is *randomly assigned* if the following conditions hold:

- $(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i$ (independence of potential outcomes and treatment).
- $0 < \Pr[D_i = 1] < 1$ (positivity).

Random assignment implies that the distribution of potential outcomes is exactly the same for the population of units that get treated and the population of units that do not. Merely observing whether or not a unit was assigned to treatment reveals nothing about its potential outcomes. Under this assumption, $E[\tau_i]$ is point identified.

Theorem 7.1.10. ATE under Random Assignment

Let $Y_i(0)$, $Y_i(1)$, and D_i be random variables with $\text{Supp}[D_i] = \{0, 1\}$. Let $Y_i = Y_i(1) \cdot D_i + Y_i(0) \cdot (1 - D_i)$ and $\tau_i = Y_i(1) - Y_i(0)$. Then, if D_i is randomly assigned,

$$E[\tau_i] = E[Y_i | D_i = 1] - E[Y_i | D_i = 0].$$

Proof: By independence (see Theorem 2.2.25, *Implications of Independence, Part III*),

$$E[Y_i(0)] = E[Y_i(0) | D_i = 0],$$

and by the potential outcomes model,

$$E[Y_i(0) | D_i = 0] = E[Y_i | D_i = 0].$$

Thus,

$$E[Y_i(0)] = E[Y_i | D_i = 0],$$

and by the same logic,

$$E[Y_i(1)] = E[Y_i(1) | D_i = 1] = E[Y_i | D_i = 1].$$

Thus,

$$\begin{aligned} E[\tau_i] &= E[Y_i(1) - Y_i(0)] \\ &= E[Y_i(1)] - E[Y_i(0)] \\ &= E[Y_i | D_i = 1] - E[Y_i | D_i = 0]. \quad \square \end{aligned}$$

Under random assignment, the plug-in estimator for the ATE is just the difference of these two sample means, which we call the *difference-in-means estimator*:

$$\hat{E}_{DM}[\tau_i] = \hat{E}[Y_i(1)] - \hat{E}[Y_i(0)] = \frac{\sum_{i=1}^n Y_i D_i}{\sum_{i=1}^n D_i} - \frac{\sum_{i=1}^n Y_i (1 - D_i)}{\sum_{i=1}^n (1 - D_i)}.$$

Much like MCAR for missing data, we can think of random assignment as allowing us, given large enough n , to impute the expected value of the unobserved potential outcomes using sample means. The following example illustrates this concept.

Example 7.1.11. *Imputation under Random Assignment*

We again consider the setting of Example 7.1.5. We have:

Unit	$Y_i(0)$	$Y_i(1)$	D_i
1	?	1	1
2	1	?	0
3	?	1	1
4	?	0	1
5	?	1	1
6	0	?	0

Assuming random assignment, $E[Y_i(1)|D_i = 0] = E[Y_i(1)]$ and $E[Y_i(0)|D_i = 1] = E[Y_i(0)]$, which we can estimate with sample means. This yields the following imputation with plug-in estimates:

Unit	$Y_i(0)$	$Y_i(1)$	D_i
1	$\hat{E}[Y_i(0)] = \frac{1}{2}$	1	1
2	1	$\hat{E}[Y_i(1)] = \frac{3}{4}$	0
3	$\hat{E}[Y_i(0)] = \frac{1}{2}$	1	1
4	$\hat{E}[Y_i(0)] = \frac{1}{2}$	0	1
5	$\hat{E}[Y_i(0)] = \frac{1}{2}$	1	1
6	0	$\hat{E}[Y_i(1)] = \frac{3}{4}$	0

Then the sample means of $Y_i(1)$ and $Y_i(0)$ with the missing values imputed are $\frac{3}{4}$ and $\frac{1}{2}$, respectively, and so our plug-in estimate of $E[\tau_i]$ is again $\frac{3}{4} - \frac{1}{2} = \frac{1}{4}$. As with MCAR, under random assignment, imputation is a redundant extra step—we are just using $\hat{E}[Y_i(1)]$ and $\hat{E}[Y_i(0)]$ to impute the missing values in order to get $\hat{E}[Y_i(1)]$ and $\hat{E}[Y_i(0)]$ again—but this way of thinking about estimating expected potential outcomes will become useful as we proceed. Δ

Even under random assignment, the only meaningful feature of τ_i that we can point identify is $E[\tau_i]$. This is because expectations are linear (Theorem 2.1.9), so we can separate $E[Y_i(1) - Y_i(0)]$ into $E[Y_i(1)] - E[Y_i(0)]$. It is not generally the case that, for an operator A , $A[Y_i(1) - Y_i(0)] = A[Y_i(1)] - A[Y_i(0)]$. The difference in medians, for instance, is not generally equal to the median difference, so we cannot identify the median treatment effect under random assignment without further assumptions. This is likewise the case for $V[\tau_i]$ and for the cumulative distribution function (CDF) of τ_i . Although the (marginal) distributions of $Y_i(1)$ and $Y_i(0)$ are identified by random assignment, their *joint* distribution, and thus the distribution of their difference, is not.

Random assignment, or a variant thereof, allows us to separate correlation from causation. Random assignment is *not* a given. Consider the

following cases, and think about why random assignment would be unlikely to hold:

Unit	Treatment	Outcome
Country	WTO Membership	GNI per Capita
Registered Voter	Voter Turnout Phone Call	Voting
City	Needle Exchange Program	HIV Prevalence

In fact, if we erroneously assume random assignment and take the difference in expected values between treated units and control units, the result can be decomposed as follows:

$$\begin{aligned} E[Y_i|D_i = 1] - E[Y_i|D_i = 0] &= E[Y_i(1) | D_i = 1] - E[Y_i(0) | D_i = 0] \\ &= E[\tau_i + Y_i(0) | D_i = 1] - E[Y_i(0) | D_i = 0] \\ &= E[\tau_i | D_i = 1] \\ &\quad + E[Y_i(0) | D_i = 1] - E[Y_i(0) | D_i = 0], \end{aligned}$$

where $E[\tau_i|D_i = 1]$ is the *average treatment effect among the treated*, and $E[Y_i(0)|D_i = 1] - E[Y_i(0)|D_i = 0]$ is selection bias.

7.1.5 Ignorability [6.1.3]

If we have additional information on each unit, we can achieve point identification of the ATE under an alternative assumption to random assignment. This assumption is known as *strong ignorability*.⁷

Definition 7.1.12. *Strong Ignorability*
Let $Y_i(0)$, $Y_i(1)$, and D_i be random variables with $\text{Supp}[D_i] = \{0, 1\}$. Let $Y_i = Y_i(1) \cdot D_i + Y_i(0) \cdot (1 - D_i)$. Then D_i is *strongly ignorable* conditional on $\mathbf{X}_i$ if the following conditions hold:

- $(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i | \mathbf{X}_i$ (conditional independence assumption).
- $\exists \epsilon > 0$ such that $\epsilon < \Pr[D_i = 1 | \mathbf{X}_i] < 1 - \epsilon$ (positivity).

(We are still assuming that treatment is binary. We will show how this assumption can be relaxed in Section 7.1.8, *Generalizing Beyond Binary Treatments*.)

⁷ Strong ignorability is named as such because it imposes a joint independence assumption $(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i | \mathbf{X}_i$. (Our definition of random assignment also invokes a joint independence assumption.) Weak ignorability only requires that $Y_i(d) \perp\!\!\!\perp D_i | \mathbf{X}_i, \forall d \in \{0, 1\}$. In theory, all our results could be obtained under weak ignorability, but the distinction is only known to be of practical relevance when considering questions of complex counterfactuals (Richardson and Robins mimeo.).

Strong ignorability is to random assignment what missing at random (MAR) is to MCAR. It is random assignment *conditional on covariates*. The conditional independence assumption states that, among units with the same measured characteristics, the types that receive the treatment and the types that do not are exactly the same in terms of their distribution of potential outcomes.

For example, if we measured age and gender, we could say that, “for people who are forty years old and male, the conditional independence assumption implies that there are no systematic differences in potential outcomes between those who get the treatment and those who do not. But we still allow for the possibility that forty-year-olds may be more or less likely to get the treatment than fifty-year-olds.” In other words, the conditional independence assumption implies that, after accounting for observable background characteristics, knowing whether or not a unit received treatment provides no additional information about a unit’s potential outcomes. This assumption is not to be taken lightly. All the concerns that we would have about the plausibility of random assignment still remain, except the burden has simply been moved to the appropriate set of *conditioning variables*. In general, we recommend only invoking strong ignorability when justifiable by substantive knowledge. Our application (Section 7.5) considers one such case.

Like random assignment, strong ignorability allows us to point identify $E[Y_i(0)]$, $E[Y_i(1)]$, and $E[\tau_i]$.

Theorem 7.1.13. ATE under Strong Ignorability

Let $Y_i(0)$, $Y_i(1)$, and D_i be random variables with $\text{Supp}[D_i] = \{0, 1\}$. Let $Y_i = Y_i(1) \cdot D_i + Y_i(0) \cdot (1 - D_i)$ and $\tau_i = Y_i(1) - Y_i(0)$, and let $\mathbf{X}_i$ be a discrete random vector.⁸ Then, if D_i is strongly ignorable conditional on $\mathbf{X}_i$,

$$\begin{aligned} E[\tau_i] &= \sum_{\mathbf{x}} E[Y_i | D_i = 1, \mathbf{X}_i = \mathbf{x}] \Pr[\mathbf{X}_i = \mathbf{x}] \\ &\quad - \sum_{\mathbf{x}} E[Y_i | D_i = 0, \mathbf{X}_i = \mathbf{x}] \Pr[\mathbf{X}_i = \mathbf{x}]. \end{aligned}$$

Proof: By the Law of Iterated Expectations,

$$E[Y_i(0)] = E\left[E[Y_i(0) | \mathbf{X}_i]\right] = \sum_{\mathbf{x}} E[Y_i(0) | \mathbf{X}_i = \mathbf{x}] \Pr[\mathbf{X}_i = \mathbf{x}],$$

and by strong ignorability (and the potential outcomes model),

$$E[Y_i(0) | \mathbf{X}_i] = E[Y_i(0) | D_i = 0, \mathbf{X}_i] = E[Y_i | D_i = 0, \mathbf{X}_i].$$

⁸ We assume here that $\mathbf{X}_i$ is discrete to ease exposition. This assumption is not necessary, but it lets us work with PMFs instead of PDFs. The continuous case is analogous. Recall that we are writing $\sum_{\mathbf{x}}$ as a shorthand for $\sum_{\mathbf{x} \in \text{Supp}[\mathbf{X}_i]}$.

Thus,

$$E[Y_i(0)] = \sum_{\mathbf{x}} E[Y_i | D_i = 0, \mathbf{X}_i = \mathbf{x}] \Pr[\mathbf{X}_i = \mathbf{x}],$$

and by the same logic,

$$E[Y_i(1)] = \sum_{\mathbf{x}} E[Y_i | D_i = 1, \mathbf{X}_i = \mathbf{x}] \Pr[\mathbf{X}_i = \mathbf{x}].$$

Thus,

$$\begin{aligned} E[\tau_i] &= E[Y_i(1)] - E[Y_i(0)] \\ &= \sum_{\mathbf{x}} E[Y_i | D_i = 1, \mathbf{X}_i = \mathbf{x}] \Pr[\mathbf{X}_i = \mathbf{x}] \\ &\quad - \sum_{\mathbf{x}} E[Y_i | D_i = 0, \mathbf{X}_i = \mathbf{x}] \Pr[\mathbf{X}_i = \mathbf{x}]. \quad \square \end{aligned}$$

Since we can observe the random vector $(Y_i, D_i, \mathbf{X}_i)$, we can point identify $E[Y_i | D_i = 1, \mathbf{X}_i = \mathbf{x}]$, $E[Y_i | D_i = 0, \mathbf{X}_i = \mathbf{x}]$, and $\Pr[\mathbf{X}_i = \mathbf{x}]$. Thus, under ignorability, $E[\tau_i]$ is point identified.

Furthermore, for every $\mathbf{x} \in \text{Supp}[\mathbf{X}_i]$, the *conditional ATE* given $\mathbf{X}_i = \mathbf{x}$ is point identified.

Theorem 7.1.14. *Conditional ATE under Strong Ignorability*

Let $Y_i(0)$, $Y_i(1)$, and D_i be random variables with $\text{Supp}[D_i] = \{0, 1\}$. Let $Y_i = Y_i(1) \cdot D_i + Y_i(0) \cdot (1 - D_i)$ and $\tau_i = Y_i(1) - Y_i(0)$, and let $\mathbf{X}_i$ be a random vector. Then, if D_i is strongly ignorable conditional on $\mathbf{X}_i$,

$$E[\tau_i | \mathbf{X}_i] = E[Y_i | D_i = 1, \mathbf{X}_i] - E[Y_i | D_i = 0, \mathbf{X}_i].$$

Proof: As in the second and third lines of the proof of Theorem 7.1.13, we have

$$E[Y_i(0) | \mathbf{X}_i] = E[Y_i(0) | D_i = 0, \mathbf{X}_i] = E[Y_i | D_i = 0, \mathbf{X}_i]$$

and

$$E[Y_i(1) | \mathbf{X}_i] = E[Y_i(1) | D_i = 1, \mathbf{X}_i] = E[Y_i | D_i = 1, \mathbf{X}_i].$$

Thus, by linearity of conditional expectations,

$$\begin{aligned} E[\tau_i | \mathbf{X}_i] &= E[Y_i(1) - Y_i(0) | \mathbf{X}_i] \\ &= E[Y_i(1) | \mathbf{X}_i] - E[Y_i(0) | \mathbf{X}_i] \\ &= E[Y_i | D_i = 1, \mathbf{X}_i] - E[Y_i | D_i = 0, \mathbf{X}_i]. \quad \square \end{aligned}$$

Note that applying the Law of Iterated Expectations to this result yields the same identification result for the ATE under ignorability as stated in Theorem 7.1.13 (*The ATE under Strong Ignorability*):

$$\begin{aligned}
 E[\tau_i] &= E[E[\tau_i | \mathbf{X}_i]] \\
 &= \sum_{\mathbf{x}} E[\tau_i | \mathbf{X}_i = \mathbf{x}] \Pr[\mathbf{X}_i = \mathbf{x}] \\
 &= \sum_{\mathbf{x}} (E[Y_i | D_i = 1, \mathbf{X}_i = \mathbf{x}] - E[Y_i | D_i = 0, \mathbf{X}_i = \mathbf{x}]) \Pr[\mathbf{X}_i = \mathbf{x}] \\
 &= \sum_{\mathbf{x}} E[Y_i | D_i = 1, \mathbf{X}_i = \mathbf{x}] \Pr[\mathbf{X}_i = \mathbf{x}] \\
 &\quad - \sum_{\mathbf{x}} E[Y_i | D_i = 0, \mathbf{X}_i = \mathbf{x}] \Pr[\mathbf{X}_i = \mathbf{x}].
 \end{aligned}$$

When both random assignment and strong ignorability with respect to a set of covariates hold, we will typically prefer to invoke strong ignorability. One key reason for this is that we can then investigate conditional ATEs, whereas we cannot with only random assignment.⁹ However, random assignment does not generally imply strong ignorability with respect to a set of covariates. In Section 7.1.7, we will discuss a case where random assignment holds, but strong ignorability with respect to a *post-treatment variable* does not.

7.1.6 The Role of the Propensity Score for Causal Inference [6.1.4]

For binary treatments, if strong ignorability holds, then instead of directly conditioning on all the covariates, it suffices to condition on a summary measure of treatment and the covariates.¹⁰ This summary measure is the *propensity score*, which corresponds to the value of the *treatment propensity function* as applied to a unit's covariate values.

Definition 7.1.15. Treatment Propensity Function

Let $Y_i(0)$, $Y_i(1)$, and D_i be random variables with $\text{Supp}[D_i] = \{0, 1\}$. Let $Y_i = Y_i(1) \cdot D_i + Y_i(0) \cdot (1 - D_i)$. Then the *treatment propensity function* is

$$p_D(\mathbf{x}) = \Pr[D_i = 1 | \mathbf{X}_i = \mathbf{x}], \forall \mathbf{x} \in \text{Supp}[\mathbf{X}_i].$$

The random variable $p_D(\mathbf{X}_i)$ is also known as the propensity score for unit i ; it is conceptually identical to the propensity score in missing data.

⁹ Another reason is that we can typically estimate causal effects more efficiently by using information on covariates. Such practice is known as *regression adjustment*; see Schochet (2010), Pitkin et al. (mimeo.), and Wager et al. (2016) for discussions.

¹⁰ This insight is attributed to Rosenbaum and Rubin (1983).

Again, the propensity score is the conditional probability of treatment given the covariates, or the CEF of D_i with respect to $\mathbf{X}_i$ evaluated at $\mathbf{x}$. The propensity score for any given covariate profile is a function of the covariates: if we know that $\mathbf{X}_i = \mathbf{x}$, and we know the full joint distribution of $(D_i, \mathbf{X}_i)$, then we know $\Pr[D_i = 1 | \mathbf{X}_i = \mathbf{x}]$.

The following theorem states the key properties of the propensity score under strong ignorability.

Theorem 7.1.16. *Strong Ignorability and the Propensity Score*

Let $Y_i(0)$, $Y_i(1)$, and D_i be random variables with $\text{Supp}[D_i] = \{0, 1\}$. Let $Y_i = Y_i(1) \cdot D_i + Y_i(0) \cdot (1 - D_i)$ and $\tau_i = Y_i(1) - Y_i(0)$, and let $\mathbf{X}_i$ be a random vector. Then, if D_i is strongly ignorable conditional on $\mathbf{X}_i$,

- $D_i \perp\!\!\!\perp \mathbf{X}_i | p_D(\mathbf{X}_i)$. (Balance conditional on $p_D(\mathbf{X}_i)$)
- $(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i | p_D(\mathbf{X}_i)$. (Conditional independence with respect to $p_D(\mathbf{X}_i)$)
- $\exists \epsilon > 0$ such that $\epsilon < \Pr[D_i = 1 | p_D(\mathbf{X}_i)] < 1 - \epsilon$. (Positivity conditional on $p_D(\mathbf{X}_i)$)

Proof: We will again prove the first two properties and omit a proof of the third; the proof follows nearly identically from Theorem 6.1.12 (*Ignorability and the Propensity Score*).

Balance: Since D_i is binary, its distribution is fully governed by its expected value (Example 5.1.2), so

$$D_i \perp\!\!\!\perp \mathbf{X}_i | p_D(\mathbf{X}_i) \iff E[D_i | \mathbf{X}_i, p_D(\mathbf{X}_i)] = E[D_i | p_D(\mathbf{X}_i)].$$

Since conditioning on a function of $\mathbf{X}_i$ provides no additional information when we have already conditioned on $\mathbf{X}_i$,

$$E[D_i | \mathbf{X}_i, p_D(\mathbf{X}_i)] = E[D_i | \mathbf{X}_i] = p_D(\mathbf{X}_i),$$

and by the Law of Iterated Expectations,

$$E[D_i | p_D(\mathbf{X}_i)] = E[E[D_i | \mathbf{X}_i] | p_D(\mathbf{X}_i)] = E[p_D(\mathbf{X}_i) | p_D(\mathbf{X}_i)] = p_D(\mathbf{X}_i).$$

Therefore,

$$E[D_i | \mathbf{X}_i, p_D(\mathbf{X}_i)] = p_D(\mathbf{X}_i) = E[D_i | p_D(\mathbf{X}_i)],$$

and thus $D_i \perp\!\!\!\perp \mathbf{X}_i | p_D(\mathbf{X}_i)$.

Conditional independence: Since D_i is binary,

$$\begin{aligned} D_i \perp\!\!\!\perp (Y_i(0), Y_i(1)) | p_D(\mathbf{X}_i) &\iff E\left[D_i \mid (Y_i(0), Y_i(1)), p_D(\mathbf{X}_i)\right] \\ &= E[D_i | p_D(\mathbf{X}_i)]. \end{aligned}$$

By the Law of Iterated Expectations,

$$\begin{aligned} & E[D_i | (Y_i(0), Y_i(1)), p_D(\mathbf{X}_i)] \\ &= E \left[E[D_i | (Y_i(0), Y_i(1)), \mathbf{X}_i] | (Y_i(0), Y_i(1)), p_D(\mathbf{X}_i) \right], \end{aligned}$$

and by strong ignorability,

$$E[D_i | (Y_i(0), Y_i(1)), \mathbf{X}_i] = E[D_i | \mathbf{X}_i] = p_D(\mathbf{X}_i).$$

So, by substitution,

$$\begin{aligned} E[D_i | (Y_i(0), Y_i(1)), p_D(\mathbf{X}_i)] &= E[p_D(\mathbf{X}_i) | (Y_i(0), Y_i(1)), p_D(\mathbf{X}_i)] \\ &= p_D(\mathbf{X}_i) \\ &= E[D_i | p_D(\mathbf{X}_i)], \end{aligned}$$

and thus $D_i \perp\!\!\!\perp (Y_i(0), Y_i(1)) | p_D(\mathbf{X}_i)$. $\square$

Again, Theorem 7.1.16 says that conditioning on the propensity score is equivalent to conditioning on all the covariates. This implies that if D_i is strongly ignorable conditional on $\mathbf{X}_i$, then D_i is also strongly ignorable conditional on the propensity score $p_D(\mathbf{X}_i)$.

It is important to note that balance on $\mathbf{X}_i$ is a *consequence* of conditioning on the propensity score. That is, after conditioning on the propensity score, the conditional distribution of covariates for treated units will be the same as the conditional distribution of covariates for control units.¹¹ In the context of observational studies, a *balance test* (a statistical test of equality of covariate distributions) merely tests the null hypothesis that we have successfully conditioned on the propensity score. If the distributions of covariates, conditional on the propensity score, for treated units and control units are significantly different, then we have done something wrong. However, balance does *not* imply that strong ignorability holds. Balance on observable characteristics does not imply balance on unobservable characteristics. Without further assumptions, there exists no general test of strong ignorability.

7.1.7 Post-Treatment Variables

In many circumstances (such as a randomized experiment), random assignment implies strong ignorability with respect to covariates. But this need not be the

¹¹ A reminder: this is not a statement about any given sample, but rather a feature of the population from which sampled units are drawn.

case: random assignment does not generally imply strong ignorability. That is, it is possible that

- $(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i$, but
- $(Y_i(0), Y_i(1)) \not\perp\!\!\!\perp D_i | X_i$.

The classic case of this is when X_i is *post-treatment*, and therefore can be affected by treatment. If D_i has a *causal effect* on X_i , then controlling for X_i can cause inconsistency, as the following example illustrates.

Example 7.1.17. *Conditioning on Post-Treatment Variables*

Suppose that the joint distribution of potential outcomes and treatment is

$$f(Y_i(0), Y_i(1), X_i(1), X_i(0), D_i) \\ = \begin{cases} \frac{1}{6} & : Y_i(0) = 0, Y_i(1) = 0, X_i(1) = 0, X_i(0) = 0, D_i = 0 \\ \frac{1}{6} & : Y_i(0) = 1, Y_i(1) = 1, X_i(1) = 1, X_i(0) = 0, D_i = 0 \\ \frac{1}{6} & : Y_i(0) = 1, Y_i(1) = 1, X_i(1) = 1, X_i(0) = 1, D_i = 0 \\ \frac{1}{6} & : Y_i(0) = 0, Y_i(1) = 0, X_i(1) = 0, X_i(0) = 0, D_i = 1 \\ \frac{1}{6} & : Y_i(0) = 1, Y_i(1) = 1, X_i(1) = 1, X_i(0) = 0, D_i = 1 \\ \frac{1}{6} & : Y_i(0) = 1, Y_i(1) = 1, X_i(1) = 1, X_i(0) = 1, D_i = 1 \\ 0 & : \text{otherwise,} \end{cases}$$

where $X_i = X_i(1)D_i + X_i(0)(1 - D_i)$. In this case, we have

- random assignment of D_i with respect to all potential outcomes: $(Y_i(0), Y_i(1), X_i(1), X_i(0)) \perp\!\!\!\perp D_i$,
- no effect of D_i on Y_i whatsoever: $Y_i(1) = Y_i(0)$ always, and
- an effect of D_i on X_i .

The distribution of observables is

$$f(Y_i, X_i, D_i) = \begin{cases} \frac{1}{6} & : Y_i = 0, X_i = 0, D_i = 0 \\ \frac{1}{6} & : Y_i = 1, X_i = 0, D_i = 0 \\ \frac{1}{6} & : Y_i = 1, X_i = 1, D_i = 0 \\ \frac{1}{6} & : Y_i = 0, X_i = 0, D_i = 1 \\ \frac{1}{6} & : Y_i = 1, X_i = 1, D_i = 1 \\ \frac{1}{6} & : Y_i = 1, X_i = 1, D_i = 1 \\ 0 & : \text{otherwise.} \end{cases}$$

Plugging into Definition 7.1.13 (*The ATE under Strong Ignorability*), we have

$$\sum_{\mathbf{x}} E[Y_i | D_i = 1, \mathbf{X}_i = \mathbf{x}] \Pr[\mathbf{X}_i = \mathbf{x}] \\ - \sum_{\mathbf{x}} E[Y_i | D_i = 0, \mathbf{X}_i = \mathbf{x}] \Pr[\mathbf{X}_i = \mathbf{x}] = -\frac{1}{4}.$$

But we know that $E[\tau_i] = 0$. Therefore conditioning on a post-treatment variable can undo even random assignment. Δ

It is generally a risky endeavor to condition on post-treatment variables, and doing so requires justifications that cannot generally be ensured even by experimental design. Note that “mediation analysis” is logically equivalent to conditioning on a post-treatment variable. Mediation analysis requires strong assumptions beyond those discussed in this chapter, and cannot be justified by random assignment of D_i alone.

7.1.8 Generalizing Beyond Binary Treatments

We have only discussed causal identification for binary treatments, where each unit either receives the treatment ($D_i = 1$) or does not ($D_i = 0$). But suppose we want to consider the causal effects of multivalued or continuous treatments, such as the effects of various amounts of cash transfers on hours worked. In this section, we demonstrate that we can generalize the NRCM to accommodate such cases. Let D_i now be any random variable.

Definition 7.1.18. Generalized Potential Outcomes Model

For an outcome variable Y_i and a treatment variable D_i , the *generalized potential outcomes model* is

$$Y_i = \left\{ Y_i(d) : D_i = d, \forall d \in \text{Supp}[D_i], \right.$$

where $Y_i(d)$ denotes the potential outcome when unit i receives treatment level d .

The generalized potential outcomes models states that, for every possible value of $d \in \text{Supp}[D_i]$, each unit i has a corresponding (stable) potential outcome $Y_i(d)$, which is observed when D_i takes on that value. When D_i is binary, this generalization reduces to the original binary potential outcomes model.

Causal effects are then defined as differences in potential outcomes for different treatment levels. For any $d, d' \in \text{Supp}[D_i]$, the causal effect of moving from d to d' is

$$\tau_i(d, d') = Y_i(d') - Y_i(d).$$

The key identification assumptions and results are all completely analogous to those we have derived for binary treatments. For example, strong ignorability would entail that

$$(Y_i(d))_{d \in \text{Supp}[D_i]} \perp\!\!\!\perp D_i | \mathbf{X}_i. \text{ (Conditional independence assumption)}$$

We will gloss over the requisite positivity assumption, since it is considerably more opaque in the generalized setting, but it basically ensures the

identifiability of all salient expected values. Theorem 7.1.13 (*The ATE under Strong Ignorability*) then generalizes as follows: under strong ignorability with discrete X_i , for any $d, d' \in \text{Supp}[D_i]$, the ATE of moving from d to d' is

$$\begin{aligned} E[\tau_i(d, d')] &= \sum_{\mathbf{x}} E[Y_i | D_i = d', X_i = \mathbf{x}] \Pr[X_i = \mathbf{x}] \\ &\quad - \sum_{\mathbf{x}} E[Y_i | D_i = d, X_i = \mathbf{x}] \Pr[X_i = \mathbf{x}]. \end{aligned}$$

The proof is completely analogous to the proof of Theorem 7.1.13. Since we can observe the random vector (Y_i, D_i, X_i) , we can point identify $E[Y_i | D_i = d', X_i = \mathbf{x}]$, $E[Y_i | D_i = d, X_i = \mathbf{x}]$, and $\Pr[X_i = \mathbf{x}]$. Thus, under ignorability, for any $d, d' \in \text{Supp}[D_i]$, $E[\tau_i(d, d')]$ is point identified. Likewise, the generalization of Theorem 7.1.14 (*The Conditional ATE under Strong Ignorability*) is: under strong ignorability, for any $d, d' \in \text{Supp}[D_i]$, the conditional average causal effect given X_i of moving from d to d' is

$$E[\tau_i(d, d') | X_i] = E[Y_i | D_i = d', X_i] - E[Y_i | D_i = d, X_i].$$

Again, the proof is completely analogous to the proof of Theorem 7.1.14. Thus, under ignorability, for any $d, d' \in \text{Supp}[D_i]$ and any $\mathbf{x} \in \text{Supp}[X_i]$, $E[\tau_i(d, d') | X_i = \mathbf{x}]$ is point identified. The Law of Iterated Expectations then implies that for any $d, d' \in \text{Supp}[D_i]$, $E[\tau_i(d, d')]$ is also point identified under ignorability, as stated above.

Given the many possible causal contrasts that we could make (d and d' are arbitrary), we might ask how to summarize the causal relationship between D_i and Y_i . The most natural summary relates back to the average partial derivative (Definition 4.3.6). If we assume that $Y_i(d)$ is differentiable with respect to d , then the following definition allows us to derive a concise measure.

Definition 7.1.19. *Average Marginal Causal Effect*

The *average marginal causal effect* of D_i is

$$AMCE = E \left[\frac{\partial E[Y_i(D_i) | D_i, X_i]}{\partial D_i} \right].$$

The Average Marginal Causal Effect is very simple to interpret: averaging over the entire distribution of D_i and X_i , how much would we expect a small change in D_i to affect Y_i ? Furthermore, under the conditional independence assumption, the CEF is causal, hence

$$AMCE = APD_{D_i} = E \left[\frac{\partial E[Y_i | D_i, X_i]}{\partial D_i} \right].$$

In other words, under strong ignorability, the average partial derivative with respect to D_i is the average marginal causal effect of D_i . In the following

section, where we consider estimation, we will not give further consideration to this generalized potential outcomes model. Generalizations to our results will largely carry through, albeit with more complexity, including working with treatment density functions instead of propensity scores.

7.2 ESTIMATION OF CAUSAL EFFECTS UNDER IGNORABILITY [6.2]

We have shown that, under random assignment, estimating $E[\tau_i]$ is straightforward: the plug-in estimator is just the difference-in-means estimator. Under strong ignorability, estimation can become much more complex. In this section, we discuss several ways to implement the strong ignorability assumption to obtain estimates of the ATE.

7.2.1 Plug-In Estimation [6.2.1]

If we have discrete covariates, then under the strong ignorability assumption, we can again estimate $E[\tau_i]$ with a simple plug-in estimator. Recall from Theorem 7.1.13 that, if $\mathbf{X}_i$ is discrete,

$$\begin{aligned} E[\tau_i] &= \sum_{\mathbf{x}} E[Y_i | D_i = 1, \mathbf{X}_i = \mathbf{x}] \Pr[\mathbf{X}_i = \mathbf{x}] \\ &\quad - \sum_{\mathbf{x}} E[Y_i | D_i = 0, \mathbf{X}_i = \mathbf{x}] \Pr[\mathbf{X}_i = \mathbf{x}] \\ &= \sum_{\mathbf{x}} (E[Y_i | D_i = 1, \mathbf{X}_i = \mathbf{x}] - E[Y_i | D_i = 0, \mathbf{X}_i = \mathbf{x}]) \Pr[\mathbf{X}_i = \mathbf{x}]. \end{aligned}$$

So the plug-in estimator is

$$\begin{aligned} \hat{E}[\tau_i] &= \sum_{\mathbf{x} \in \widehat{\text{Supp}}[\mathbf{X}_i]} (\hat{E}[Y_i | D_i = 1, \mathbf{X}_i = \mathbf{x}] - \hat{E}[Y_i | D_i = 0, \mathbf{X}_i = \mathbf{x}]) \hat{\Pr}[\mathbf{X}_i = \mathbf{x}] \\ &= \sum_{\mathbf{x} \in \widehat{\text{Supp}}[\mathbf{X}_i]} \left(\frac{\sum_{i=1}^n Y_i D_i \cdot \mathbf{I}(\mathbf{X}_i = \mathbf{x})}{\sum_{i=1}^n D_i \cdot \mathbf{I}(\mathbf{X}_i = \mathbf{x})} - \frac{\sum_{i=1}^n Y_i (1 - D_i) \cdot \mathbf{I}(\mathbf{X}_i = \mathbf{x})}{\sum_{i=1}^n (1 - D_i) \cdot \mathbf{I}(\mathbf{X}_i = \mathbf{x})} \right) \\ &\quad \cdot \frac{\sum_{i=1}^n \mathbf{I}(\mathbf{X}_i = \mathbf{x})}{n}, \end{aligned}$$

where $\mathbf{I}(\cdot)$ is the indicator function and $\widehat{\text{Supp}}[\mathbf{X}_i]$ is the set of all observed values of $\mathbf{X}_i$.¹² This estimator is sometimes known as the post-stratification estimator for causal inference, which is named after its analog in the context

¹² This requires that we have at least one treated unit and one control unit with $\mathbf{X}_i = \mathbf{x}$ for every covariate profile $\mathbf{x} \in \widehat{\text{Supp}}[\mathbf{X}_i]$, otherwise we will have a zero in the denominator. See Footnote 15.

of missing data. It can be seen as a weighted average of the difference-in-means estimators for units with the same covariate values, where the weights are the observed frequencies of those covariate profiles.

Example 7.2.1. Plug-In Estimation of the ATE

We return to Example 7.1.5 (*Causal Inference as a Missing Data Problem*), but now assume that we also observe a binary covariate X_i for every unit:

Unit	$Y_i(0)$	$Y_i(1)$	D_i	X_i
1	?	1	1	1
2	1	?	0	0
3	?	1	1	0
4	?	0	1	1
5	?	1	1	1
6	0	?	0	1

The above plug-in estimator assuming ignorability yields the estimate

$$\begin{aligned}\hat{E}[\tau_i] &= \left(\hat{E}[Y_i | D_i = 1, X_i = 0] - \hat{E}[Y_i | D_i = 0, X_i = 0] \right) \widehat{\Pr}[X_i = 0] \\ &\quad + \left(\hat{E}[Y_i | D_i = 1, X_i = 1] - \hat{E}[Y_i | D_i = 0, X_i = 1] \right) \widehat{\Pr}[X_i = 1] \\ &= (1 - 1) \cdot \frac{2}{6} + \left(\frac{2}{3} - 0 \right) \cdot \frac{4}{6} \\ &= \frac{4}{9}.\end{aligned}$$

Note that this estimate differs from the random assignment-based estimate of $\frac{1}{4}$.

Like random assignment, we can think of strong ignorability as allowing us to impute the unobserved potential outcomes, only now we impute the missing potential outcome for each unit using the sample mean of units in the other treatment condition *with the same covariate values*. In Example 7.1.5, this yields

Unit	$Y_i(0)$	$Y_i(1)$	D_i	X_i
1	$\hat{E}[Y_i(0) X_i = 1] = 0$	1	1	1
2	1	$\hat{E}[Y_i(1) X_i = 0] = 1$	0	0
3	$\hat{E}[Y_i(0) X_i = 0] = 1$	1	1	0
4	$\hat{E}[Y_i(0) X_i = 1] = 0$	0	1	1
5	$\hat{E}[Y_i(0) X_i = 1] = 0$	1	1	1
6	0	$\hat{E}[Y_i(1) X_i = 1] = \frac{2}{3}$	0	1

Then the sample means of $Y_i(1)$ and $Y_i(0)$ with the missing values imputed are $\frac{7}{9}$ and $\frac{1}{3}$, respectively, and so our estimate of $E[\tau_i]$ is again $\frac{7}{9} - \frac{1}{3} = \frac{4}{9}$. Δ

Plug-in estimation of this type works when we have only a small number of possible covariate values. In such cases, the estimator is consistent under the usual plug-in regularity conditions. When we have many covariates, or continuous covariates, simple plug-in estimation generally is not feasible. In the remainder of this chapter, we discuss some other ways of estimating the ATE under strong ignorability.

7.2.2 Regression Estimation [6.2.2]

Recall Theorem 7.1.14: under strong ignorability, the conditional ATE given $\mathbf{X}_i$ is point identified as

$$E[\tau_i|\mathbf{X}_i] = E[Y_i|D_i = 1, \mathbf{X}_i] - E[Y_i|D_i = 0, \mathbf{X}_i].$$

Under strong ignorability, the difference between $E[Y_i|D_i = 1, \mathbf{X}_i]$ and $E[Y_i|D_i = 0, \mathbf{X}_i]$ is equivalent to a causal effect. But notice that $E[Y_i|D_i, \mathbf{X}_i]$ is a conditional expectation function (CEF). When strong ignorability holds, some features of the CEF of Y_i are imbued with a causal interpretation.

We have a readily available tool for estimating the CEF: ordinary least squares (OLS) regression consistently estimates the best linear predictor (BLP) of the CEF—where “best” means minimum mean squared error (MSE). Suppose we assumed that, at least to a first approximation,

$$E[Y_i|D_i, \mathbf{X}_i] = \beta_0 + \beta_1 D_i + \mathbf{X}_i \boldsymbol{\beta}.$$

Then

$$\begin{aligned} E[\tau_i|\mathbf{X}_i] &= E[Y_i|D_i = 1, \mathbf{X}_i] - E[Y_i|D_i = 0, \mathbf{X}_i] \\ &= \beta_0 + \beta_1 \cdot 1 + \mathbf{X}_i \boldsymbol{\beta} - (\beta_0 + \beta_1 \cdot 0 + \mathbf{X}_i \boldsymbol{\beta}) \\ &= \beta_1, \end{aligned}$$

so we can estimate the conditional ATE given $\mathbf{X}_i$ using the OLS coefficient estimate $\hat{\beta}_1$:

$$\hat{E}[\tau_i|\mathbf{X}_i] = \hat{\beta}_1.$$

Then, by the Law of Iterated Expectations,

$$\begin{aligned} E[\tau_i] &= \sum_{\mathbf{x}} E[\tau_i|\mathbf{X}_i = \mathbf{x}] \Pr[\mathbf{X}_i = \mathbf{x}] \\ &= \sum_{\mathbf{x}} \beta_1 \Pr[\mathbf{X}_i = \mathbf{x}] \\ &= \beta_1 \sum_{\mathbf{x}} \Pr[\mathbf{X}_i = \mathbf{x}] \\ &= \beta_1 \cdot 1 \\ &= \beta_1, \end{aligned}$$

so we can likewise estimate the ATE with $\hat{E}[\tau_i] = \hat{\beta}_1$. Thus, insofar as $E[Y_i|D_i, \mathbf{X}_i] \approx \beta_0 + \beta_1 D_i + \mathbf{X}_i \boldsymbol{\beta}$, we can interpret $\hat{\beta}_1$ as a good approximation of the ATE. This is illustrated in the following example.

Example 7.2.2. *Causal Inference with OLS Regression*
Suppose we had the following data:

Unit	$Y_i(0)$	$Y_i(1)$	D_i	$X_{[1]i}$	$X_{[2]i}$
1	?	2	1	1	7
2	5	?	0	8	2
3	?	3	1	10	3
4	?	10	1	3	1
5	?	2	1	5	2
6	0	?	0	7	0

We want to fill in all the missing values with $\hat{E}[Y_i|D_i = d, X_{[1]i} = x_{[1]}, X_{[2]i} = x_{[2]}]$. We can see how we could do this using regression. We will start by assuming that, at least to a first approximation,

$$E[Y_i|D_i, X_{[1]i}, X_{[2]i}] = \beta_0 + \beta_1 D_i + \beta_2 X_{[1]i} + \beta_3 X_{[2]i}.$$

Then we could estimate the coefficients with OLS and use the resulting equation to impute the missing values:

Unit	$Y_i(0)$	$Y_i(1)$	D_i	$X_{[1]i}$	$X_{[2]i}$
1	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 1 + \hat{\beta}_3 \cdot 7$	2	1	1	7
2	5	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 \cdot 8 + \hat{\beta}_3 \cdot 2$	0	8	2
3	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 9 + \hat{\beta}_3 \cdot 3$	3	1	9	3
4	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 3 + \hat{\beta}_3 \cdot 1$	10	1	3	1
5	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 5 + \hat{\beta}_3 \cdot 2$	2	1	5	2
6	0	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 \cdot 7 + \hat{\beta}_3 \cdot 0$	0	7	0

(Remember: these are just expectations for units with the same covariate values; we cannot actually fill in the missing potential outcomes.) Equivalently, we could instead impute all the potential outcomes, which will yield the same estimate for $E[\tau_i]$ and is more straightforward to implement:

Unit	$Y_i(0)$	$Y_i(1)$	D_i	$X_{[1]i}$	$X_{[2]i}$
1	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 1 + \hat{\beta}_3 \cdot 7$	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 \cdot 1 + \hat{\beta}_3 \cdot 7$	1	1	7
2	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 8 + \hat{\beta}_3 \cdot 2$	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 \cdot 8 + \hat{\beta}_3 \cdot 2$	0	8	2
3	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 9 + \hat{\beta}_3 \cdot 3$	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 \cdot 9 + \hat{\beta}_3 \cdot 3$	1	9	3
4	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 3 + \hat{\beta}_3 \cdot 1$	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 \cdot 3 + \hat{\beta}_3 \cdot 1$	1	3	1
5	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 5 + \hat{\beta}_3 \cdot 2$	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 \cdot 5 + \hat{\beta}_3 \cdot 2$	1	5	2
6	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 7 + \hat{\beta}_3 \cdot 0$	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 \cdot 7 + \hat{\beta}_3 \cdot 0$	0	7	0

In either case, the estimate of $E[\tau_i]$ is just the difference-in-means estimator with the imputed values. This estimator would be consistent if ignorability

held and if the functional form of the CEF were in fact $E[Y_i|D_i, X_{[1]i}, X_{[2]i}] = \beta_0 + \beta_1 D_i + \beta_2 X_{[1]i} + \beta_3 X_{[2]i}$. In the case without interactions or polynomials, the resulting estimate of $E[\tau_i]$ is just $\hat{\beta}_1$. $\triangle$

When n is small, we are likely to want to fit a simple linear approximation to avoid overfitting (Section 4.3.2). Our data are too micronumerous to fit a more complicated model. In larger samples, we could use polynomials, interactions or any other way of accounting for nonlinearity discussed in Section 4.3. One particularly common approach—closely mirroring the approach from missing data—is to fit separate (or *unpooled*) regression models across the treatment and control groups. This specification is logically equivalent to running a single regression model with D_i interacted with all elements of $\mathbf{X}_i$, and yields consistent estimates when $E[Y_i|D_i = d, \mathbf{X}_i] = \beta_{d0} + \mathbf{X}_i \boldsymbol{\beta}_d$, $\forall d \in \{0, 1\}$.

In general, to quickly produce an estimate of the ATE following regression estimation, it is easiest to take predicted values for all potential outcomes and then take the difference in means. The logic of this procedure is clear from the plug-in principle:

$$E[\tau_i] = E[Y_i(1)] - E[Y_i(0)] = E[E[Y_i|D_i = 1, \mathbf{X}_i]] - E[E[Y_i|D_i = 0, \mathbf{X}_i]],$$

for which a natural plug-in estimator is the difference in means of predicted values. We refer to this estimator as the regression estimator for causal inference.

Definition 7.2.3. *Regression Estimator for Causal Inference*

Let $Y_i(0)$, $Y_i(1)$, and D_i be random variables with $\text{Supp}[D_i] = \{0, 1\}$. Let $Y_i = Y_i(1) \cdot D_i + Y_i(0) \cdot (1 - D_i)$ and $\tau_i = Y_i(1) - Y_i(0)$, and let $\mathbf{X}_i$ be a random vector. Then, given n i.i.d. observations of $(Y_i, D_i, \mathbf{X}_i)$, the *regression estimator* for $E[\tau_i]$ is

$$\begin{aligned} \hat{E}_R[\tau_i] &= \hat{E}[Y_i(1)] - \hat{E}[Y_i(0)] \\ &= \frac{1}{n} \sum_{i=1}^n \hat{E}[Y_i|D_i = 1, \mathbf{X}_i] - \frac{1}{n} \sum_{i=1}^n \hat{E}[Y_i|D_i = 0, \mathbf{X}_i], \end{aligned}$$

where $\hat{E}[Y_i|D_i = d, \mathbf{X}_i = \mathbf{x}]$ is an estimator of the CEF.

As the regression estimator is a plug-in estimator, we have a good sense of its theoretical properties: the estimator will behave well when the CEF is well characterized by the specification of the regression and the usual plug-in regularity conditions hold.

In this section, we have established conditions for using regression to obtain principled estimates of causal relationships. This result is at the heart of the difference between our approach to causal inference and the standard treatment

in econometrics and statistical modeling. We have established a basis for causal inference with regression *without*

- making any assumptions about an error distribution (such as homoskedasticity and normality),
- assuming linearity (except as an approximation that can be made arbitrarily flexible), or
- discussing “exogeneity,” “endogeneity,” “omitted variables bias,” or “errors uncorrelated with regressors.”

We have invoked a *substantive* assumption about the relationship between potential outcomes and treatment. This assumption is strong, but we need not conflate this essential identification assumption with the details of statistical specification.

7.2.3 Maximum Likelihood Plug-In Estimation of Causal Effects

Using the tools developed in Section 5.2.4 (*Plug-In Estimation and Conditional Expectation Functions*), it is possible to directly wed the causal identification assumptions here to the parametric models discussed in Chapter 5. Suppose that we have strong ignorability and further suppose that we have a parametric model that is true: $g(y, (d, \mathbf{x}), \boldsymbol{\theta}) = f_{Y_i|(D_i, \mathbf{X}_i)}(y|(d, \mathbf{x}))$. Then it is the case that all average potential outcomes over the support of D_i are point identified, and can be written in terms of the parametric model. The average potential outcome when $D_i = d$ is

$$\begin{aligned} E[Y_i(d)] &= \int_{-\infty}^{\infty} E[Y_i|D_i = d, \mathbf{X}_i = \mathbf{x}] f_{\mathbf{X}_i}(\mathbf{x}) d\mathbf{x} \\ &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} yg(y, (d, \mathbf{x}), \boldsymbol{\theta}) dy \right) f_{\mathbf{X}_i}(\mathbf{x}) d\mathbf{x}, \end{aligned}$$

where the first equality holds due to ignorability, and the second holds due to proper specification of the parametric model.

Recall that Definition 7.2.3 (*Regression Estimator for Causal Inference*) does not exclude estimates of the CEF produced by other means. Thus, combining Definition 5.2.10 (*Maximum Likelihood Plug-In Estimator*) and Definition 7.2.3, we can see that the maximum likelihood plug-in estimator for $E[Y_i(d)]$ is computed by substituting $\hat{\boldsymbol{\theta}}_{ML}$ for $\boldsymbol{\theta}$ and replacing the expected value with respect to $\mathbf{X}_i$ with a sample mean:

$$\hat{E}[Y_i(d)] = \frac{1}{n} \sum_{i=1}^n \left(\int_{-\infty}^{\infty} yg(y, (d, \mathbf{x}), \hat{\boldsymbol{\theta}}_{ML}) dy \right).$$

The ML plug-in estimator for $E[\tau_i]$ is therefore given by

$$\hat{E}_{ML}[\tau_i] = \frac{1}{n} \sum_{i=1}^n \left(\int_{-\infty}^{\infty} yg(y, (1, \mathbf{X}_i), \hat{\theta}_{ML}) dy - \int_{-\infty}^{\infty} yg(y, (0, \mathbf{X}_i), \hat{\theta}_{ML}) dy \right).$$

If the parametric model is correct, and the usual plug-in regularity conditions hold, maximum likelihood plug-in estimation works as usual here.

7.2.4 Matching [6.2.3]

The most common, but certainly not only, propensity score-based method for causal inference is *propensity score matching*, an estimator analogous to hot-deck imputation. There are many types of matching, but we will discuss just one type: one-to-one, nearest-neighbor, with-replacement, propensity score matching. For the moment, we will ignore the need to estimate propensity scores and assume that they are known. (We consider estimation of propensity scores in the following section.)

Example 7.2.4. Propensity Score Matching

Suppose we had the following data:

Unit	$Y_i(0)$	$Y_i(1)$	D_i	$X_{[1]i}$	$X_{[2]i}$	$p_D(\mathbf{X}_i)$
1	?	2	1	1	7	0.33
2	5	?	0	8	2	0.14
3	?	3	1	10	3	0.73
4	?	10	1	3	1	0.35
5	?	2	1	5	2	0.78
6	0	?	0	7	0	0.70

Assume that strong ignorability holds and that our goal is to estimate $E[\tau_i]$. By Theorem 7.1.16 (*Strong Ignorability and the Propensity Score*), strong ignorability and the potential outcomes model imply that

$$\begin{aligned} E[Y_i(0) \mid D_i = 1, p_D(\mathbf{X}_i)] &= E[Y_i(0) \mid D_i = 0, p_D(\mathbf{X}_i)] \\ &= E[Y_i \mid D_i = 0, p_D(\mathbf{X}_i)] \end{aligned}$$

and

$$\begin{aligned} E[Y_i(1) \mid D_i = 0, p_D(\mathbf{X}_i)] &= E[Y_i(1) \mid D_i = 1, p_D(\mathbf{X}_i)] \\ &= E[Y_i \mid D_i = 1, p_D(\mathbf{X}_i)], \end{aligned}$$

so we can impute the missing potential outcomes with estimates of $E[Y_i \mid D_i = d, p_D(\mathbf{X}_i)]$:

Unit	$Y_i(0)$	$Y_i(1)$	D_i	$X_{[1]i}$	$X_{[2]i}$	$p_D(\mathbf{X}_i)$
1	$\hat{E}[Y_i D_i = 0, p_D(\mathbf{X}_i) = 0.33]$	2	1	1	7	0.33
2	5	$\hat{E}[Y_i D_i = 1, p_D(\mathbf{X}_i) = 0.14]$	0	8	2	0.14
3	$\hat{E}[Y_i D_i = 0, p_D(\mathbf{X}_i) = 0.73]$	3	1	10	3	0.73
4	$\hat{E}[Y_i D_i = 0, p_D(\mathbf{X}_i) = 0.35]$	10	1	3	1	0.35
5	$\hat{E}[Y_i D_i = 0, p_D(\mathbf{X}_i) = 0.78]$	2	1	5	2	0.78
6	0	$\hat{E}[Y_i D_i = 1, p_D(\mathbf{X}_i) = 0.70]$	0	7	0	0.70

Then, if our estimator of $E[Y_i|D_i = d, p_D(\mathbf{X}_i) = p_D(\mathbf{x})]$ is consistent, the difference in means with the missing potential outcomes imputed will be a consistent estimator of the ATE.

There are many imputation-type estimators for causal inference that exploit ignorability. (Regression is itself an imputation estimator.) The procedure for (one-to-one, nearest-neighbor) matching is perhaps the simplest possible: for each treated unit, find the control unit that is closest on $p_D(\mathbf{X}_i)$ and use that unit’s outcome to fill in the missing control potential outcome for the treated unit. Likewise, for each control unit, find the treated unit that is closest on $p_D(\mathbf{X}_i)$ and use that unit’s outcome to fill in the missing treatment potential outcome for the control unit.

In this example, this yields

Unit	$Y_i(0)$	$Y_i(1)$	D_i	$X_{[1]i}$	$X_{[2]i}$	$p_D(\mathbf{X}_i)$
1	5	2	1	1	7	0.33
2	5	2	0	8	2	0.14
3	0	3	1	10	3	0.73
4	5	10	1	3	1	0.35
5	0	2	1	5	2	0.78
6	0	3	0	7	0	0.70

So the matching estimate of $E[\tau_i]$ is $\frac{22}{6} - \frac{15}{6} = \frac{7}{6} \approx 1.17$. Δ

All matching estimators are based on some variant of this logic. Some versions match on some other summary metric of $\mathbf{X}_i$. Some do one-to- k matching. Some choose observations to empirically maximize a measure of the balance of the matched groups. Some even choose, whenever possible, observations that *exactly* match on all covariates—though of course this requires, at minimum, that $\mathbf{X}_i$ be discrete.

Why is one-to-one, nearest-neighbor matching not such a great idea, even when we assume strong ignorability? Why would we not want to just use the outcome for the closest unit in the other treatment condition to estimate a missing potential outcome? Intuitively, matching only uses a small amount of the information available to impute the missing potential outcomes. In the above example, we used $Y_1(1) = 2$ to impute $Y_2(1)$ because $P_1 = 0.33$ is closest to $P_2 = 0.14$. But $P_4 = 0.35$ is almost as close to P_1 , so $Y_4(1) = 10$ probably

also gives us some information about $E[Y_i(1)|p_D(\mathbf{X}_i) = 0.14]$. So why would we ignore $Y_4(1)$ entirely and just use $Y_1(1)$ as our guess for $Y_2(1)$?

The problems with this method are exacerbated when the treatment propensity function is unknown. In the following section, we consider estimation of propensity scores. However, matching estimators are *not* plug-in estimators, and so their properties do not operate quite as expected. For example, even if we have a consistent estimator of $p_D(\mathbf{X}_i)$, matching with estimated propensity scores is not generally root- n consistent—the bias (and therefore MSE) of the estimator shrinks “too slowly” as n grows large—and, thus, the usual methods for constructing confidence intervals, *including the bootstrap*, do not generally work.¹³

7.2.5 Maximum Likelihood Plug-In Estimation of Propensity Scores [6.2.4]

Estimation of propensity scores is unchanged from missing data. We reiterate the discussion here. Recall from Definition 7.1.15 that the propensity score $p_D(\mathbf{x}) = \Pr[D_i = 1|\mathbf{X}_i = \mathbf{x}]$, and, from Example 2.1.3, that $\Pr[D_i = 1|\mathbf{X}_i = \mathbf{x}] = E[D_i|\mathbf{X}_i = \mathbf{x}]$. Therefore, estimating the treatment propensity function is equivalent to estimating the CEF of D_i with respect to $\mathbf{X}_i$. Just as with missing data, this is a case where OLS would not be suitable for estimating an unknown CEF. A linear approximation could predict propensity score estimates that are greater than one or less than zero. A natural path forward lies with a binary choice model (Section 5.1.2) that respects the constraints on the distribution of D_i .

Suppose now that we have a binary choice model for treatment with mean function h . If this model is true, then following Section 5.2.4, we can use a ML estimator for the propensity score, $\hat{p}_D(\mathbf{x}) = h(\mathbf{x}\hat{\boldsymbol{\beta}}_{ML})$. If the binary choice model is true, then, under the usual plug-in regularity conditions, $\hat{p}_D(\mathbf{x})$ will be consistent for $p_D(\mathbf{x})$. As with missing data, if the model is not true, then our propensity score estimates will only be as good as our approximation permits. Again, with large n , we can use polynomials, interactions, mixture models, or other relaxations of the model in order to obtain a better approximation. Regardless of the exact specification, we will assume as we proceed that we are using an ML plug-in estimate from a binary choice model, $\hat{p}_D(\mathbf{x})$, for our plug-in estimates, though our results do not depend on this.

7.2.6 Weighting Estimators [6.2.5]

Matching is not the only way to use the propensity score to estimate $E[\tau_i]$. Much as they do in the case of missing data, inverse probability-weighted (IPW) estimators provide one such method. Before we can describe these estimators, we first establish an important population result. Transforming

¹³ See, e.g., Abadie and Imbens (2008).

the outcomes by weighting by $\frac{1}{p_D(\mathbf{X}_i)}$ (if in treatment) or $\frac{1}{1-p_D(\mathbf{X}_i)}$ (if in control) allows us to write the ATE as a simple expected value.

Theorem 7.2.5. *Reweighting under Strong Ignorability*

Let $Y_i(0)$, $Y_i(1)$, and D_i be random variables with $\text{Supp}[D_i] = \{0, 1\}$. Let $Y_i = Y_i(1) \cdot D_i + Y_i(0) \cdot (1 - D_i)$ and $\tau_i = Y_i(1) - Y_i(0)$, and let $\mathbf{X}_i$ be a random vector. Then, if D_i is strongly ignorable conditional on $\mathbf{X}_i$,

$$E[\tau_i] = E\left[\frac{Y_i D_i}{p_D(\mathbf{X}_i)} - \frac{Y_i(1 - D_i)}{1 - p_D(\mathbf{X}_i)}\right].$$

Proof: By definition,

$$E\left[\frac{Y_i D_i}{p_D(\mathbf{X}_i)} \mid \mathbf{X}_i\right] = E\left[Y_i \cdot \frac{D_i}{\Pr[D_i = 1 \mid \mathbf{X}_i]} \mid \mathbf{X}_i\right],$$

and by the potential outcomes model,

$$E\left[Y_i \cdot \frac{D_i}{\Pr[D_i = 1 \mid \mathbf{X}_i]} \mid \mathbf{X}_i\right] = E\left[Y_i(1) \cdot \frac{D_i}{\Pr[D_i = 1 \mid \mathbf{X}_i]} \mid \mathbf{X}_i\right].$$

Then, since strong ignorability holds, $Y_i(1) \perp\!\!\!\perp D_i \mid \mathbf{X}_i$, so

$$\begin{aligned} E\left[Y_i(1) \cdot \frac{D_i}{\Pr[D_i = 1 \mid \mathbf{X}_i]} \mid \mathbf{X}_i\right] &= E[Y_i(1) \mid \mathbf{X}_i] \cdot E\left[\frac{D_i}{\Pr[D_i = 1 \mid \mathbf{X}_i]} \mid \mathbf{X}_i\right] \\ &= E[Y_i(1) \mid \mathbf{X}_i] \cdot \frac{E[D_i \mid \mathbf{X}_i]}{\Pr[D_i = 1 \mid \mathbf{X}_i]} \\ &= E[Y_i(1) \mid \mathbf{X}_i] \cdot \frac{\Pr[D_i = 1 \mid \mathbf{X}_i]}{\Pr[D_i = 1 \mid \mathbf{X}_i]} \\ &= E[Y_i(1) \mid \mathbf{X}_i]. \end{aligned}$$

Thus,

$$E\left[\frac{Y_i D_i}{p_D(\mathbf{X}_i)} \mid \mathbf{X}_i\right] = E[Y_i(1) \mid \mathbf{X}_i].$$

So, by the Law of Iterated Expectations,

$$E[Y_i(1)] = E[E[Y_i(1) \mid \mathbf{X}_i]] = E\left[E\left[\frac{Y_i D_i}{p_D(\mathbf{X}_i)} \mid \mathbf{X}_i\right]\right] = E\left[\frac{Y_i D_i}{p_D(\mathbf{X}_i)}\right],$$

and by the same logic,

$$E[Y_i(0)] = E[E[Y_i(0) \mid \mathbf{X}_i]] = E\left[E\left[\frac{Y_i(1 - D_i)}{1 - p_D(\mathbf{X}_i)} \mid \mathbf{X}_i\right]\right] = E\left[\frac{Y_i(1 - D_i)}{1 - p_D(\mathbf{X}_i)}\right].$$

Thus, by linearity of expectations,

$$\begin{aligned} E[\tau_i] &= E[Y_i(1) - Y_i(0)] \\ &= E[Y_i(1)] - E[Y_i(0)] \\ &= E\left[\frac{Y_i D_i}{p_D(\mathbf{X}_i)}\right] - E\left[\frac{Y_i(1 - D_i)}{1 - p_D(\mathbf{X}_i)}\right] \\ &= E\left[\frac{Y_i D_i}{p_D(\mathbf{X}_i)} - \frac{Y_i(1 - D_i)}{1 - p_D(\mathbf{X}_i)}\right]. \quad \square \end{aligned}$$

The IPW estimator is again a plug-in estimator: we replace $p_D(\mathbf{X}_i)$ with an estimate thereof, $\hat{p}_D(\mathbf{X}_i)$ (as in the previous section), and substitute sample means for expected values.

Definition 7.2.6. *Inverse Probability-Weighted (IPW) Estimator for Causal Inference*

Let $Y_i(0)$, $Y_i(1)$, and D_i be random variables with $\text{Supp}[D_i] = \{0, 1\}$. Let $Y_i = Y_i(1) \cdot D_i + Y_i(0) \cdot (1 - D_i)$ and $\tau_i = Y_i(1) - Y_i(0)$, and let $\mathbf{X}_i$ be a random vector. Then, given n i.i.d. observations of $(Y_i, D_i, \mathbf{X}_i)$, the *inverse probability-weighted estimator* for $E[\tau_i]$ is

$$\hat{E}_{IPW}[\tau_i] = \frac{1}{n} \sum_{i=1}^n \left(\frac{Y_i D_i}{\hat{p}_D(\mathbf{X}_i)} - \frac{Y_i(1 - D_i)}{1 - \hat{p}_D(\mathbf{X}_i)} \right).$$

The IPW estimator weights every treated unit's outcome by the inverse of its probability of being in treatment and every control unit's outcome by the inverse of its probability of being in control, and then takes the difference in means (with missing potential outcomes effectively set equal to zero).

The logic behind IPW estimators for causal inference is the same as for missing data. Suppose that some units are very unlikely to be assigned to treatment, so they have small propensity scores. These units are likely to be underrepresented in the treatment group and overrepresented in the control group, so we want to up-weight them when they are in treatment and down-weight them when they are in control. The reverse goes for units that are unlikely to be assigned to control: we want to up-weight them when they are in control and down-weight them when they are in treatment. In the case where we have a discrete and finite number of values for $\mathbf{X}_i$, the IPW estimator is logically equivalent to the plug-in estimator from Section 7.2.1 (*Plug-In Estimation*).

The IPW estimator has high variability (that is, its standard error is large) with small samples. An alternative that performs better in practice is the causal inference version of the stabilized IPW estimator.

Definition 7.2.7. *Stabilized IPW Estimator for Causal Inference*

Let $Y_i(0)$, $Y_i(1)$, and D_i be random variables with $\text{Supp}[D_i] = \{0, 1\}$. Let $Y_i = Y_i(1) \cdot D_i + Y_i(0) \cdot (1 - D_i)$ and $\tau_i = Y_i(1) - Y_i(0)$, and let $\mathbf{X}_i$ be a random vector. Then, given n i.i.d. observations of $(Y_i, D_i, \mathbf{X}_i)$, the *stabilized IPW estimator* for $E[\tau_i]$ is

$$\hat{E}_{SIPW}[\tau_i] = \frac{\frac{1}{n} \sum_{i=1}^n \frac{Y_i D_i}{\hat{p}_D(\mathbf{X}_i)}}{\frac{1}{n} \sum_{i=1}^n \frac{D_i}{\hat{p}_D(\mathbf{X}_i)}} - \frac{\frac{1}{n} \sum_{i=1}^n \frac{Y_i(1 - D_i)}{1 - \hat{p}_D(\mathbf{X}_i)}}{\frac{1}{n} \sum_{i=1}^n \frac{1 - D_i}{1 - \hat{p}_D(\mathbf{X}_i)}}.$$

When $\hat{p}_D(\mathbf{X}_i)$ converges (in an appropriate functional sense) to $p_D(\mathbf{X}_i)$, then the denominator of each fraction will converge in probability to one. To see this, note that, with the true propensity score, $E[\frac{D_i}{p_D(\mathbf{X}_i)}] = E[\frac{1-D_i}{1-p_D(\mathbf{X}_i)}] = 1$. However, this normalization provides benefits in estimation. In the case where we draw an unusually large number of units where $p_D(\mathbf{X}_i)$ is large or small, the stabilized IPW estimator's denominators adjust accordingly. Because the stabilized IPW estimator accounts for this between-sample variation in $\sum_{i=1}^n \frac{1}{p_D(\mathbf{X}_i)}$ and $\sum_{i=1}^n \frac{1}{1-p_D(\mathbf{X}_i)}$, it typically confers efficiency gains over the IPW estimator in practice. Valid asymptotic inference for IPW estimators can be achieved under strong ignorability, proper specification of $\hat{p}_D(\mathbf{X}_i)$, and the usual plug-in regularity conditions via the bootstrap, re-estimating the propensity scores in each bootstrap sample.

7.2.7 Doubly Robust Estimators [6.2.6]

Finally, as with missing data, it is also possible to combine regression with weighting specifications using a doubly robust (DR) estimator. All our results from the case of missing data carry through naturally. Under strong ignorability, these approaches yield consistent estimates when either the regression specification or the propensity score specification is correct. Before we state this result, we note an important population result.

Theorem 7.2.8. *Double Robustness Theorem for Causal Inference*

Let $Y_i(0)$, $Y_i(1)$, and D_i be random variables with $\text{Supp}[D_i] = \{0, 1\}$. Let $Y_i = Y_i(1) \cdot D_i + Y_i(0) \cdot (1 - D_i)$ and $\tau_i = Y_i(1) - Y_i(0)$, and let $\mathbf{X}_i$ be a random vector. Let $\tilde{E}[Y_i | D_i = d, \mathbf{X}_i = \mathbf{x}]$ be an approximation of the CEF evaluated at $D_i = d$ and $\mathbf{X}_i = \mathbf{x}$, and let $\tilde{p}_D(\mathbf{X}_i)$ be an approximation of the propensity score. If strong ignorability holds and if either

- (i) $\tilde{E}[Y_i|D_i = d, \mathbf{X}_i = \mathbf{x}] = E[Y_i|D_i = d, \mathbf{X}_i = \mathbf{x}], \forall d \in \{0, 1\}, \mathbf{x} \in \text{Supp}[\mathbf{X}_i]$,
and $\exists \epsilon > 0$ such that $\epsilon < \tilde{p}_D(\mathbf{x}) < 1 - \epsilon$,

or

- (ii) $\tilde{p}_D(\mathbf{x}) = p_D(\mathbf{x}), \forall \mathbf{x} \in \text{Supp}[\mathbf{X}_i]$,

then

$$E[\tau_i] = E \left[\tilde{E}[Y_i|D_i = 1, \mathbf{X}_i] + \frac{D_i(Y_i - \tilde{E}[Y_i|D_i = 1, \mathbf{X}_i])}{\tilde{p}_D(\mathbf{X}_i)} - \tilde{E}[Y_i|D_i = 0, \mathbf{X}_i] - \frac{(1 - D_i)(Y_i - \tilde{E}[Y_i|D_i = 0, \mathbf{X}_i])}{1 - \tilde{p}_D(\mathbf{X}_i)} \right].$$

Proof: We will show that

$$E[Y_i(1)] = E \left[\tilde{E}[Y_i|D_i = 1, \mathbf{X}_i] + \frac{D_i(Y_i - \tilde{E}[Y_i|D_i = 1, \mathbf{X}_i])}{\tilde{p}_D(\mathbf{X}_i)} \right].$$

To consider case (i), suppose that $\tilde{E}[Y_i|D_i = d, \mathbf{X}_i = \mathbf{x}] = E[Y_i|D_i = d, \mathbf{X}_i = \mathbf{x}], \forall d \in \{0, 1\}, \mathbf{x} \in \text{Supp}[\mathbf{X}_i]$. Then we have

$$\begin{aligned} & E \left[\tilde{E}[Y_i|D_i = 1, \mathbf{X}_i] + \frac{D_i(Y_i - \tilde{E}[Y_i|D_i = 1, \mathbf{X}_i])}{\tilde{p}_D(\mathbf{X}_i)} \right] \\ &= E \left[E[Y_i|D_i = 1, \mathbf{X}_i] + \frac{D_i(Y_i - E[Y_i|D_i = 1, \mathbf{X}_i])}{\tilde{p}_D(\mathbf{X}_i)} \right] \\ &= E[E[Y_i|D_i = 1, \mathbf{X}_i]] + E \left[\frac{D_i Y_i}{\tilde{p}_D(\mathbf{X}_i)} \right] - E \left[\frac{D_i E[Y_i|D_i = 1, \mathbf{X}_i]}{\tilde{p}_D(\mathbf{X}_i)} \right] \\ &= E[E[Y_i|D_i = 1, \mathbf{X}_i]] + E \left[\frac{D_i E[Y_i|D_i = 1, \mathbf{X}_i]}{\tilde{p}_D(\mathbf{X}_i)} \right] \\ &\quad - E \left[\frac{D_i E[Y_i|D_i = 1, \mathbf{X}_i]}{\tilde{p}_D(\mathbf{X}_i)} \right] \\ &= E[E[Y_i|D_i = 1, \mathbf{X}_i]] \\ &= E[Y_i(1)], \end{aligned}$$

where the third equality follows from the Law of Iterated Expectations and the fifth follows from strong ignorability (see Theorem 7.1.13).

To consider case (ii), suppose instead that $\tilde{p}_D(\mathbf{x}) = p_D(\mathbf{x}), \forall \mathbf{x} \in \text{Supp}[\mathbf{X}_i]$. Then

$$\begin{aligned}
 & \mathbb{E} \left[\tilde{\mathbb{E}}[Y_i | D_i = 1, \mathbf{X}_i] + \frac{D_i(Y_i - \tilde{\mathbb{E}}[Y_i | D_i = 1, \mathbf{X}_i])}{\tilde{p}_D(\mathbf{X}_i)} \right] \\
 &= \mathbb{E} \left[\tilde{\mathbb{E}}[Y_i | D_i = 1, \mathbf{X}_i] + \frac{D_i(Y_i - \tilde{\mathbb{E}}[Y_i | D_i = 1, \mathbf{X}_i])}{p_D(\mathbf{X}_i)} \right] \\
 &= \mathbb{E} \left[\tilde{\mathbb{E}}[Y_i | D_i = 1, \mathbf{X}_i] \right] + \mathbb{E} \left[\frac{D_i Y_i}{p_D(\mathbf{X}_i)} \right] - \mathbb{E} \left[\frac{D_i \tilde{\mathbb{E}}[Y_i | D_i = 1, \mathbf{X}_i]}{p_D(\mathbf{X}_i)} \right] \\
 &= \mathbb{E} \left[\tilde{\mathbb{E}}[Y_i | D_i = 1, \mathbf{X}_i] \right] + \mathbb{E} \left[\frac{D_i Y_i}{p_D(\mathbf{X}_i)} \right] - \mathbb{E} \left[\tilde{\mathbb{E}}[Y_i | D_i = 1, \mathbf{X}_i] \right] \\
 &= \mathbb{E} \left[\frac{D_i Y_i}{p_D(\mathbf{X}_i)} \right] \\
 &= \mathbb{E}[Y_i(1)].
 \end{aligned}$$

All results carry through analogously to $\mathbb{E}[Y_i(0)]$, and linearity of expectations completes the result. $\square$

As with missing data, the Double Robustness Theorem establishes that if either the approximation of the CEF or the approximation of the treatment propensity function are exactly correct, then we can write the average treatment effect in terms of simple expected values of the observable data and the approximations. This result motivates a plug-in estimator, obtained by replacing expected values with sample means, $\tilde{\mathbb{E}}[Y_i | D_i = d, \mathbf{X}_i = \mathbf{x}]$ with some estimate $\hat{\mathbb{E}}[Y_i | D_i = d, \mathbf{X}_i = \mathbf{x}]$, and $\hat{p}_D(\mathbf{X}_i)$ with some estimate $\hat{p}_D(\mathbf{X}_i)$.

Definition 7.2.9. *Doubly Robust (DR) Estimator for Causal Inference*

Let $Y_i(0)$, $Y_i(1)$, and D_i be random variables with $\text{Supp}[D_i] = \{0, 1\}$. Let $Y_i = Y_i(1) \cdot D_i + Y_i(0) \cdot (1 - D_i)$ and $\tau_i = Y_i(1) - Y_i(0)$, and let $\mathbf{X}_i$ be a random vector. Then, given n i.i.d. observations of $(Y_i, D_i, \mathbf{X}_i)$, the *doubly robust estimator* is

$$\begin{aligned}
 \hat{\mathbb{E}}_{DR}[\tau_i] &= \frac{1}{n} \sum_{i=1}^n \hat{\mathbb{E}}[Y_i | D_i = 1, \mathbf{X}_i] + \frac{1}{n} \sum_{i=1}^n \frac{D_i(Y_i - \hat{\mathbb{E}}[Y_i | D_i = 1, \mathbf{X}_i])}{\hat{p}_D(\mathbf{X}_i)} \\
 &\quad - \frac{1}{n} \sum_{i=1}^n \hat{\mathbb{E}}[Y_i | D_i = 0, \mathbf{X}_i] - \frac{1}{n} \sum_{i=1}^n \frac{(1 - D_i)(Y_i - \hat{\mathbb{E}}[Y_i | D_i = 0, \mathbf{X}_i])}{1 - \hat{p}_D(\mathbf{X}_i)},
 \end{aligned}$$

where, $\forall d \in \{0, 1\}$, $\hat{\mathbb{E}}[Y_i | D_i = d, \mathbf{X}_i]$ is an estimator for $\mathbb{E}[Y_i | D_i = d, \mathbf{X}_i]$, and $\hat{p}_D(\mathbf{X}_i)$ is an estimator for $p_D(\mathbf{X}_i)$.

The DR estimator can be decomposed much as in the case of missing data. The first term, $\frac{1}{n} \sum_{i=1}^n \hat{E}[Y_i | D_i = 1, \mathbf{X}_i]$, is a standard regression estimator from Section 7.2.2 (*Regression Estimation*). The second term, $\frac{1}{n} \sum_{i=1}^n \frac{D_i(Y_i - \hat{E}[Y_i | D_i = 1, \mathbf{X}_i])}{\hat{p}_D(\mathbf{X}_i)}$, is an IPW estimator correcting for any unusual deviations in the actual data from the regression estimate. Analogous calculations carry through the latter two sums. As a plug-in estimator, if the conditions for Theorem 7.2.8 and the usual plug-in regularity conditions hold, the DR estimator can be used in the usual manner, with the bootstrap available for inference. The DR estimator often provides a practical solution to the statistical problems posed by implementing the strong ignorability assumption.

7.2.8 Placebo Testing

Readers may ask: are there any ways to *test* strong ignorability? The answer is a (somewhat qualified) no: although some assumptions can be tested, the conditional independence assumption cannot directly be tested. Because of the Fundamental Problem of Causal Inference, we can *never* observe whether the distributions of potential outcomes for treated units are different from those of control units (conditional on covariates).

However, depending on the substance of the study in question, it is sometimes possible to present statistical evidence that may strengthen the credibility of the research design. Suppose that we have an additional set of *placebo* (or *negative control*) variables $\mathbf{W}_i$ that we do not think we need to condition on and that are not causally affected by the treatment assignment.

Then we can test the null hypothesis that

$$\mathbf{W}_i \perp\!\!\!\perp D_i | \mathbf{X}_i.$$

This is known as a *placebo test*.¹⁴ One simple way to implement this is by performing an *F*-test. An *F*-test compares a regression of D_i on $\mathbf{W}_i$ and $\mathbf{X}_i$ with a regression of D_i on $\mathbf{X}_i$ alone, and asks: does adding $\mathbf{W}_i$ to the regression explain more of the remaining variation in D_i than we would expect by chance? If so, then we can reject the null hypothesis that $\mathbf{W}_i$ is independent of D_i conditional on $\mathbf{X}_i$.

Substantively, we may not be willing to believe that $(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i | \mathbf{X}_i$ holds if it is statistically unlikely that $\mathbf{W}_i \perp\!\!\!\perp D_i | \mathbf{X}_i$ holds. That is, if we can show that, even conditional on $\mathbf{X}_i$, treated units are systematically different from control units in at least some ways, then we should worry that they might also be systematically different in terms of their potential outcomes. In

¹⁴ Placebo tests of this type should not be confused with balance tests after matching or inverse probability weighting. Balance tests are used to assess whether or not we have successfully implemented the matching or weighting algorithm. By Theorem 7.1.16 (*Strong Ignorability and the Propensity Score*), if we have the right propensity score specification, we know that the “balancing property” ensures that we will have statistical independence of D_i and $\mathbf{X}_i$.

such cases, imbalance on $\mathbf{W}_i$ provides valuable evidence about the validity of our identifying assumptions.

7.3 OVERLAP AND POSITIVITY

In the previous section, we discussed placebo tests designed to assess the plausibility of the conditional independence assumption, $(Y_i(0), Y_i(1)) \perp D_i | \mathbf{X}_i$. Another important issue arises relating to the positivity assumption, $\exists \epsilon > 0$ such that $\epsilon < p_D(\mathbf{X}_i) < 1 - \epsilon$. In this section, we consider the consequences of the positivity assumption not holding; for instance, what if $\exists \mathbf{x} \in \text{Supp}[\mathbf{X}_i]$ such that $p_D(\mathbf{x}) = 0$ or $p_D(\mathbf{x}) = 1$.¹⁵

In practice, even if the positivity assumption holds, but $\exists \mathbf{x} \in \text{Supp}[\mathbf{X}_i]$ such that $p_D(\mathbf{x})$ is near zero or one, then difficulties in estimation may ensue. Both of these issues are tied to the idea of “overlap” in the distributions of $\mathbf{X}_i$ across treatment and control arms. We proceed by illustrating that a failure of the positivity assumption logically implies incomplete (or “limited”) overlap in the population of interest. When the positivity assumption fails, then the distributions of $\mathbf{X}_i$ for treated and control units have regions with positive probability that do not completely overlap with one another. This notion of overlap is formalized in the following definition.

Definition 7.3.1. *Complete Population Overlap*

Let D_i be a random variable with $\text{Supp}[D_i] = \{0, 1\}$, and let $\mathbf{X}_i$ be a random vector. Then D_i has *complete population overlap* with respect to $\mathbf{X}_i$ if

$$\text{Supp}[\mathbf{X}_i | D_i = 1] = \text{Supp}[\mathbf{X}_i | D_i = 0].$$

Overlap simply implies that, if there exists a covariate value in the treatment group with positive (conditional) density, then that covariate value also has positive density conditional on being the control group, and vice versa. We now state our primary result in this section, which establishes that positivity and overlap are tightly connected.

Theorem 7.3.2. *Positivity Implies Complete Population Overlap*

Let D_i be a random variable with $\text{Supp}[D_i] = \{0, 1\}$, and let $\mathbf{X}_i$ be a random vector. If $\exists \epsilon > 0$ such that $\epsilon < p_D(\mathbf{X}_i) < 1 - \epsilon$, then D_i has complete population overlap with respect to $\mathbf{X}_i$.

Furthermore, the converse of this claim (that population overlap implies positivity) can almost be stated here, but requires some measure-theoretic

¹⁵ In the analogous problem of missing data, we might worry that $\exists \mathbf{x} \in \text{Supp}[\mathbf{X}_i]$ such that $p_R(\mathbf{x}) = 0$. Our results in this section apply analogously to Chapter 6.

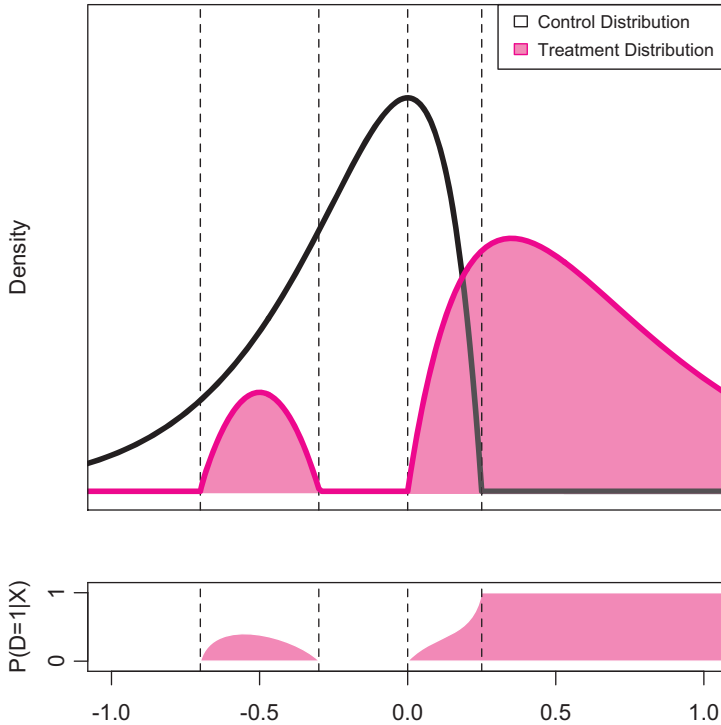


FIGURE 7.3.1. *Limited Population Overlap Illustration*

conditions on support. We omit the formal proof, but provide an illustrative example in the case of a single covariate.

Example 7.3.3. Limited Population Overlap

The upper part of Figure 7.3.1 graphs the probability density functions of a single covariate X_i conditional on $D_i = 1$ (in red) and $D_i = 0$ (in black); population overlap fails to hold in the ranges of X_i where one of the conditional densities is positive but the other is zero. The lower part of the figure graphs the propensity score $p_D(X_i) = \Pr[D_i = 1 | X_i]$; positivity fails to hold in the ranges of X_i where the propensity score is zero or one. Note that the failures of overlap and positivity occur in exactly the same regions: in the upper range of X_i , where there are treated units but no control units, the propensity score must equal one, and in the ranges where there are control units but no treated units, the propensity score must equal zero. Δ

When overlap fails to hold, there are values of X_i where the conditional independence assumption is not sufficient for identification of $E[Y_i(1) - Y_i(0) | X_i]$, since either all such units are treated or all such units are untreated. Therefore, the population average treatment effect $E[Y_i(1) - Y_i(0)]$ is not

identified without additional assumptions. Approaches to causal inference in this situation include bounding exercises, extrapolation from other regions of the data, and changing the target population. We next consider the latter of these approaches, which can often be done agnostically while retaining practical applicability.

7.3.1 Changing the Target Population

A problem of limited overlap is not just mechanical, but rather may also reflect a more fundamental problem. For instance, we may be interested in the question of the effects of Medicare takeup on a given health outcome; but how might we consider the impact of Medicare takeup for non-Americans—or, on non-seniors, for that matter? Would it make sense to talk about the treatment effects of Medicare takeup for such populations? The fact that zero non-Americans and zero non-seniors take up Medicare may lead us to question whether some queries even admit meaningful answers. In such circumstances, and even in less straightforward cases, the researcher may wish to consider examining an alternative population that features variation in treatment assignments along all salient dimensions.

We could even define alternative populations from the population under study, and condition on the types of observations that have some probability of having variation in $\mathbf{X}_i$. In such cases, it might make sense to restrict our attention to, say, $E[Y_i(d) | \epsilon < p_D(\mathbf{X}_i) < 1 - \epsilon]$, for some researcher-specified value of $\epsilon \in (0, .5)$. As long as $\Pr[\epsilon < p_D(\mathbf{X}_i) < 1 - \epsilon] > 0$, positivity holds by definition for this subpopulation. We might also choose an estimand that weights members of the population by some function of the propensity score. For example, the estimand $E[p_D(\mathbf{X}_i)\tau_i]$, which weights by the propensity score itself, is equivalent to the *average treatment effect on the treated (ATT)*, $E[\tau_i | D_i = 1]$. The ATT is nonparametrically identified under the CIA and a weaker positivity assumption, $p_D(\mathbf{X}_i) < \epsilon < 1$. In the case of the Medicare example, the ATT would correspond to the average effect of Medicare takeup among those who took up Medicare. Other targets are also possible. Angrist and Krueger (1999) demonstrates that a regression of Y_i on D_i and $p_D(\mathbf{X}_i)$ recovers a weighted average treatment effect with weights $p_D(\mathbf{X}_i)(1 - p_D(\mathbf{X}_i))$. Thus, if the usual plug-in regularity conditions hold, the treatment propensity function is known (or can be estimated consistently), and the CIA holds, then regression will consistently estimate the average treatment effect for a reweighted population. Such actions resolve technical issues of positivity, but perhaps introduce nuance in the interpretation that is not immediately evident to the researcher (Crump et al. 2006).

7.3.2 Empirical Overlap, Micronumerosity, and Weighting Estimators

As usual, our final topic is data. If we could know whether or not positivity held, we could adjust our target accordingly. But in finite samples, it might

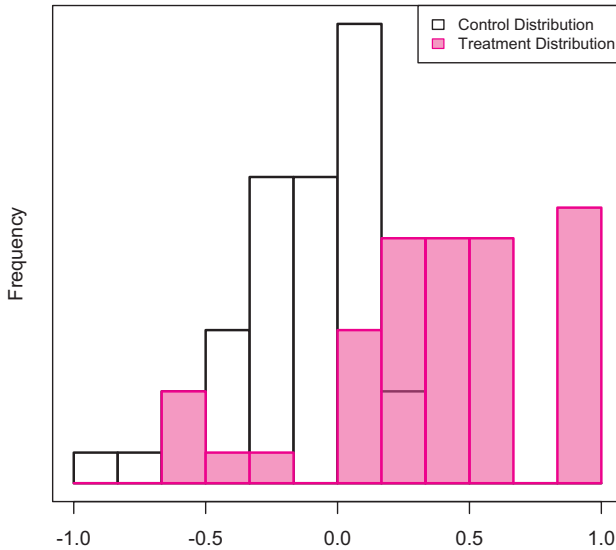


FIGURE 7.3.2. *Empirical Overlap Illustration*

be difficult to distinguish between a failure of the positivity assumption or just not having enough data. We will show that this problem is again one of micronumerosity. To see this, we consider the empirical analog to the example in Section 7.3.

Example 7.3.4. *Limited Empirical Overlap*

Our data here are a random sample from the distribution illustrated in Figure 7.3.1, and we have presented a histogram in Figure 7.3.2 that represents the observed distributions of X_i in the treatment group and the control group. Consider the region where $X_i < -0.67$. Our sample has some control units but no treated units in this range. From the observed data alone, we cannot tell whether the problem is a failure of positivity or just micronumerosity (in which case we would expect to eventually see some treated units with $X_i < -0.67$ if we increased the sample size). But either way, if additional data collection is not an option, then the lack of empirical overlap in certain regions (such as $X_i < -0.67$ and $X_i > 0.33$) implies that we cannot rely on the conditional independence assumption alone for estimation of the population average treatment effect; such endeavors must lean on other approximating assumptions. $\triangle$

These concerns can be fundamental unless we are willing to impose more structure on the problem. Suppose, for example, that we have a propensity score model that we believe to be accurate. Then we can use it to change

the target population, either by trimming based on the empirical values, or, more generally, by selecting an alternative target population through reweighting. This might have good properties, particularly if the researcher is skeptical about the validity of the positivity assumption for their target population.

If the substance of the problem necessitates inference on the full population, and we believe that positivity does in fact hold, then a failure of empirical overlap is simply a case of micronumerosity. In order to estimate the ATE when n is small and overlap is limited, we may have to use an estimator that relies on a restrictive specification (such as linearity of the CEF). Ultimately, however, our best hope for obtaining a good approximation of the ATE lies in the collection of more data. To fully appreciate this tie to micronumerosity, we can note that propensity scores near zero or one have extremely negative consequences for the behavior of the IPW estimator,

$$\hat{\tau}_{IPW}[\tau_i] = \frac{1}{n} \sum_{i=1}^n \left(\frac{Y_i D_i}{p_D(\mathbf{X}_i)} - \frac{Y_i(1 - D_i)}{1 - p_D(\mathbf{X}_i)} \right).$$

Intuitively, when $p_D(\mathbf{X}_i)$ is near zero or one, we can see that extremely low or extremely high values can be obtained if we happen to have a “chance” draw of one or more units with extremely large IPW weights. (The stabilized IPW and DR estimators do much to control these problems, but nevertheless the disproportionate influence that observations with high IPW weights will have increases the variability of our estimators.) The problem can be exacerbated in small samples when the propensity scores are estimated, leading to an additional way in which the IPW weights might become very large. This motivates a practice known as *Winsorization*, which is the practice that we have used and will be using in our applications for our weighting-type estimators, including DR estimators.¹⁶

Definition 7.3.5. *Winsorized (Treatment) Propensity Score*

Suppose we have a (treatment) propensity score estimate $\hat{p}_D(\mathbf{x})$ for some $\mathbf{x}$. Then, given a *Winsorization* $\epsilon \in (0, .5)$, the associated *Winsorized estimate* is

$$\hat{p}_{D,W}(\mathbf{x}) = \begin{cases} 1 - \epsilon & : \hat{p}_D(\mathbf{x}) \geq 1 - \epsilon \\ \hat{p}_D(\mathbf{x}) & : \epsilon < \hat{p}_D(\mathbf{x}) < 1 - \epsilon \\ \epsilon & : \hat{p}_D(\mathbf{x}) \leq \epsilon. \end{cases}$$

For example, if we set $\epsilon = 0.01$, then our propensity score estimates are constrained to lie in $[0.01, 0.99]$. Winsorization thus controls the sampling variance of our weighting estimators, as it prevents chance events from wildly

¹⁶ In the case of missing data, we only need to Winsorize so that the Winsorized estimate $\hat{p}_{R,W}$ is constrained to be above ϵ .

distorting the distribution of our estimator. But just as in sieve estimation, we need not take ϵ as fixed. As data permit, we can admit estimated propensity scores that increasingly tend toward zero or one. If positivity does indeed hold, then Winsorization will become an exceedingly rare event as n grows large.

7.4 FURTHER EXTENSIONS

We note that our treatment of causal inference is incomplete with respect to the myriad ways that point identification of causal parameters can be achieved. At their core, however, most such approaches for causal inference rely on an assumption resembling ignorability. We consider some examples. Instrumental variable strategies typically require an ignorable *instrument*, which is a variable that affects the outcome only through the channel of one particular treatment. Regression discontinuity designs typically require an assumption closely resembling ignorability at a single point in some *forcing variable* that determines whether or not a unit is treated. Approaches that attempt to use panel (longitudinal) data to achieve point identification usually depend on some type of ignorability assumption, though there are many that can be invoked: ignorability conditional on unit (*fixed effects* estimation), ignorability conditional on behavior and treatments in prior periods (*sequential ignorability*), or an assumption that treatment is ignorable with respect to trends in potential outcomes (*difference-in-difference* estimation).

The key ideas expressed in this chapter are readily extendable to most identification strategies commonly used in the social and health sciences. As usual, regression, weighting, or DR estimators can be used to implement these assumptions. There is no magic in achieving causal inference; there is simply the union of substantive assumptions and statistical approximation. Sometimes these approaches are sufficient to achieve consistent point estimates of the quantities of interest, but a healthy dose of agnosticism goes a long way.

7.5 APPLICATION: THE EFFECT OF GENDER ON SWISS CITIZENSHIP APPROVAL VOTES

We consider an application using data from Hainmueller and Hangartner (2013). Hainmueller and Hangartner examine Swiss municipalities where naturalization requests were decided at the municipality level by referendum. Voters in these municipalities received leaflets with detailed descriptions of immigrant applicants, and then cast secret ballots to vote “yes” or “no” on each individual’s request. Applicants receiving a majority of “yes” votes were approved for citizenship. This dataset covers 2,419 naturalization referenda held between 1977 and 2003 in forty-four Swiss municipalities, so $n = 2,419$.¹⁷

¹⁷ The original dataset contained 2,429 observations; we exclude from analysis 10 observations for which the age of the applicant was unknown. As all 10 of these observations were

The outcome of interest is percentage of “no” votes on the referendum, and we examine whether gender has a causal effect on the percentage of “no” votes. Gender is coded as one if male and zero if female. Of the 2,419 applicants, 1,640 identified as male and 779 as female.¹⁸ Covariates are region of origin, age, years since arrival, education, unemployment period, refugee status, marriage status, language, level of integration, number of applications, attractiveness in photograph, skill of labor, and decade of evaluation.¹⁹ We assume the data are i.i.d. draws of (Y_i, D_i, X_i) , where Y_i denotes percentage of “no” votes on an applicant’s referendum, D_i is a binary variable for gender, and X_i is the covariate matrix. Our goal is to estimate the effect that the described gender of the applicant has on the outcome.

The counterfactual interpretation is interesting here, and interpreting the potential outcomes model— $Y_i = Y_i(1) \cdot D_i + Y_i(0) \cdot (1 - D_i)$ —relies on substantive judgment. Gender is not typically a trait that can easily be viewed as causal—it is embedded deeply in identity, and its meaning is complex. However, we can identify a very particular form of counterfactual: what if the applicant description had been *exactly* the same, except the gender of the applicant had been switched? This setting resembles experiments where researchers send out otherwise identical job or housing applications for evaluation, but experimentally manipulate the gender (see, e.g., Moss-Racusin et al. 2012). Here, the presented gender is not experimentally manipulated, but our potential outcome interpretation remains the same. Thus, when we define potential

male applicants, simple plug-in type estimates fail in this setting: our plug-in estimate of the propensity score will be one for all such subjects.

¹⁸ No applicants identified with a non-binary gender.

¹⁹ Region of origin reports the country or region the applicant has come from: Northern and Western Europe, Southern Europe, Central and Eastern Europe, the former Yugoslavia, Turkey, Asian countries, and other non-European countries. Age is recorded as one for only one of the following indicators, and zero otherwise: if the applicant is 21–40 years old, 41–60 years old, and over 60 years old. Years since arrival is recorded as number of years from the applicant’s arrival in Switzerland to time of application, divided by 10. Education is measured as the number of years of formal education. Unemployment period is coded as one if the description of the applicant mentions any period of unemployment and zero otherwise. Refugee status is coded as one if the applicant entered Switzerland as a refugee and zero otherwise. Marriage status is coded as one if the applicant is married and zero otherwise. Language measures the applicant’s Swiss German skills, as reported on the leaflet, as “excellent,” “good,” “sufficient,” or “insufficient.” Level of integration is reported under several variables based on how applicants were described to voters: “assimilated” is coded as quasi-continuous with scores between 0 and 2; “integrated” is coded as quasi-continuous with scores between 0 and 2; and “adjusted,” “indistinguishable from a Swiss citizen,” and “familiar with Swiss customs and traditions” are all coded as binary variables. Number of applications measures the number of times the applicant has applied for naturalization. Attractiveness in photograph was coded by Hainmueller and Hangartner’s research assistants as “attractive” or “average.” Skill of labor aggregates ISCO-88 codes, coding the first two groups as highly skilled, groups three to five as medium skilled, and the rest as low skilled. Decade of evaluation reports the decade of the referendum. Missing covariate values are addressed with indicator variables in analysis.

outcomes, $Y_i(0)$ refers to the “no” vote share we would (or did) observe had the applicant description been exactly as it was, but with the gender identified as female; and $Y_i(1)$ refers to the “no” vote share we would (or did) observe had the applicant description been exactly as it was, but with the gender identified as male.

Assuming the potential outcomes model—and the fact that vote shares are bounded between 0 and 100%—we can compute bounds on the average potential outcome $E[Y_i(0)]$, that is, the average “no” vote share had each application had a female-identified applicant. We can of course also compute bounds on the ATE $E[\tau_i] = E[Y_i(1) - Y_i(0)]$, that is, the ATE of switching each application from female-identified to male-identified, all else equal. The bounds in this setting are quite wide, to the point of being largely uninformative. Even without accounting for uncertainty, our point estimates imply that the effect of gender could be anywhere between -54.156 and 45.844 percentage points. Without further assumptions, this is the best we can do.

We turn now to assumptions that allow us to point identify average potential outcomes and treatment effects. In this setting, the random assignment assumption is unlikely. There are many reasons to expect that the applications of men may differ from the applications of women in ways that would affect the judgments of voters, and thus $(Y_i(0), Y_i(1)) \not\perp D_i$. For example, men and women may have different education levels, marital statuses, or employment statuses, perhaps due to long-term structural or cultural influences. If we were to assume random assignment and use the plug-in estimator (difference-in-means), we have a point estimate of $\hat{E}[\tau_i] = 1.335$, with a two-tailed p -value (against the null hypothesis that $E[\tau_i] = 0$) of 0.051 —very nearly reaching the conventional significance level of $p = 0.05$.

The fact that random assignment is unlikely to hold does not mean that all hope is lost. This is a setting where the ignorability assumption, implying $(Y_i(0), Y_i(1)) \perp D_i | \mathbf{X}_i$, is quite plausible. As long as voters do not directly know the applicant, the applicant description contains *all* of the information available to voters. Assuming that the researchers’ coding of the applicant descriptions—that is, the variables in $\mathbf{X}_i$ —does indeed account for all the information presented to the voters, then, *by definition*, D_i must be conditionally independent of potential outcomes.

If we assume ignorability, then we again have a variety of estimators available. We begin by using two variants of regression estimation to impute missing potential outcomes. The first, OLS (linear), uses OLS under the working assumption that the CEF of the observed data is linear in D_i and $\mathbf{X}_i$: $E[Y_i | D_i, \mathbf{X}_i] \approx \beta_0 + \beta_1 D_i + \mathbf{X}_i \boldsymbol{\beta}$. The second, OLS (unpooled), fits separate regression models to be fitted for male and female applicants: $E[Y_i | D_i = d, \mathbf{X}_i] \approx \beta_{d0} + \mathbf{X}_i \boldsymbol{\beta}_d$, $\forall d \in \{0, 1\}$. The estimated ATEs from these two specifications largely agree in direction and magnitude, yielding respective ATE estimates of 0.706 and 1.146 . We see that conditioning on $\mathbf{X}_i$ has reduced the

magnitude of our estimated effect, and the estimates are no longer near traditional levels of statistical significance, as the p -value against the null hypothesis of zero ATE is 0.165.

Again, in order to use the propensity score-based estimators, we need to first estimate the propensity score, $p_D(\mathbf{X}_i) = \Pr[D_i = 1 | \mathbf{X}_i]$. Again, we will impose the working assumption that $\Pr[D_i = 1 | \mathbf{X}_i] \approx \frac{e^{\mathbf{X}_i \boldsymbol{\gamma}}}{(1 + e^{\mathbf{X}_i \boldsymbol{\gamma}})}$ and use maximum likelihood (ML) estimation to estimate each $\hat{p}_D(\mathbf{X}_i)$. (See Sections 5.1.2, *Binary Choice Model* and 5.2, *Maximum Likelihood Estimation*.) Immediately, we see that one-to-one, nearest-neighbor matching appears to perform abysmally. We estimate that its standard error is nearly 4 times as large as that associated with the OLS (linear) specification, and the associated point estimate of $E[\tau_i]$ is half again as large as the random assignment plug-in estimator.²⁰ This is likely to be attributable to the pathologies of matching-type estimators discussed in Section 7.2.4 (*Matching*).

We turn now to the IPW and DR estimators.²¹ The IPW estimator has markedly bad performance, with the second-largest estimated standard error, and an estimate of $E[\tau_i]$ nearly three times as large as the random assignment plug-in estimator. The stabilized IPW estimator, on the other hand, yields the smallest of our estimated treatment effects, with $\hat{E}[\tau_i] = 0.316$. The fact that the IPW estimates are in such disagreement with the other estimates and with each other may suggest some misspecification of the propensity score model. For the DR estimator, we combine the OLS (unpooled) specification for imputation with the propensity score specification for the IPW estimators.²² The DR estimator largely agrees with the OLS-based imputation estimate, albeit with some loss of precision, perhaps attributable to potential propensity score misspecification.

Table 7.5.1 summarizes these estimates. With the exception of the (potentially pathological) matching and IPW estimators, the ignorability-based estimates reflect a downward adjustment from the random assignment-based plug-in estimate. (The OLS specifications largely agree with their more technical cousin: the DR estimator.) If we accept the ignorability assumption, then, substantively, this suggests that men have traits that are, on average, less desirable to the voting public. After conditioning on their observable differences from women, we find no estimated effect of gender that approaches traditional levels of statistical significance, though our estimates still suggest that—all

²⁰ As with hot deck imputation, the bootstrap may not work as a method for computing valid confidence intervals for matching estimators since the bias of the estimator may converge very slowly (see, e.g., Abadie and Imbens 2008), but the bootstrap standard errors are likely to adequately characterize sampling variability.

²¹ As discussed in Section 7.3.2, we Winsorize our propensity score estimates to lie within $[0.01, 0.99]$.

²² It is standard practice for DR estimators to rely on unpooled regression fits across treatment arms for imputation.

TABLE 7.5.1. *Estimates for Effect of Gender on Percentage “No” Votes*

Estimator	Target	Estimate	SE
Lower Bound Plug-In	$E[Y_i(0)]$	11.523	0.388
	$E[\tau_i]$	−54.156	0.429
Upper Bound Plug-In	$E[Y_i(0)]$	79.319	0.657
	$E[\tau_i]$	45.844	0.429
Random Assignment	$E[Y_i(0)]$	35.780	0.557
	$E[\tau_i]$	1.335	0.685
OLS (Linear)	$E[Y_i(0)]$	36.206	0.487
	$E[\tau_i]$	0.706	0.508
OLS (Unpooled)	$E[Y_i(0)]$	35.700	0.649
	$E[\tau_i]$	1.146	0.646
Matching	$E[Y_i(0)]$	34.732	1.951
	$E[\tau_i]$	2.009	1.988
IPW	$E[Y_i(0)]$	33.690	0.997
	$E[\tau_i]$	3.731	1.088
Stabilized IPW	$E[Y_i(0)]$	36.638	0.756
	$E[\tau_i]$	0.316	0.753
Doubly Robust	$E[Y_i(0)]$	35.676	0.701
	$E[\tau_i]$	1.187	0.709

Estimates are of average potential outcome for female applicants ($E[Y(0)]$) and ATE of male identification ($E[\tau]$) with respect to “no” vote share in referenda on approving Swiss citizenship applications (Hainmueller and Hangartner 2013).

else equal—being male decreases voter support for one’s immigration application. Together, our results highlight how a clever observational study can shed light on well-defined causal quantities and illustrate the importance of wedding credible identifying assumptions to sensible estimation strategies.

7.6 FURTHER READINGS

As might be expected given the parallel structures of Chapter 6 and Chapter 7, the readings from Chapter 6 remain highly relevant here. Angrist and Pischke (2009), Morgan and Winship (2014), and Hernan and Robins (in press) provide the most natural extensions from our materials here and are highly recommended; they discuss practical strategies for obtaining point identification for causal quantities. Angrist and Krueger (1999) contains an excellent and highly practical article covering of many of the same materials as Angrist and Pischke (2009). Manski (1990) provides a now-canonical derivation of the bounds that we discussed in Section 7.1.3, along with assumptions that may reduce the width of these bounds. VanderWeele (2015) provides a book-length

discussion of strategies for identification of complex counterfactual quantities, including mediation effects and spillover effects. Rubin (2006) provides a book-length discussion of matching-based methods. There are also three review articles on developments in causal inference that we recommend: the first two are Imbens and Wooldridge (2009) and Athey and Imbens (mimeo.), which each discuss causal identification strategies and associated estimators. But perhaps most importantly, Richardson and Rotnitzky (2014) contains an excellent review of the enormously influential and wide-ranging body of work by James Robins. We cannot recommend this review article highly enough for those seeking a starting point to explore more advanced topics in causal inference, particularly with respect to longitudinal data.

As with Chapter 6, we did not consider design-based approaches to causal inference. One strain of this research, Fisherian randomization inference, focuses on testing sharp null hypotheses (that is, hypotheses where the causal effect for every unit is hypothesized, rather than just average treatment effects) and constructing confidence intervals accordingly. This approach has excellent book-length treatments in Rosenbaum (2002) and in the more introductory Rosenbaum (2010). Another type of design-based research follows from Splawa-Neyman, Dabrowska, and Speed (1923), wherein the inferential targets are average treatment effects defined on finite populations. This line of research was repopularized by Robins (1988), Reichardt and Gollob (1999), Freedman (2008a), and Freedman (2008b). We strongly recommend these articles, along with the extraordinarily clear and elegant response to Freedman given by Lin (2013). We further recommend Aronow and Samii (in press), which provides a generalization of this framework for generic experimental designs and permits some forms of interference between units. Imbens and Rubin (2011) is a valuable textbook that includes a significant focus on Neyman-style design-based causal inference.

Finally, we reiterate that while we did not consider graph-based approaches for identification of causal quantities in this book, we view the topic as highly important. Pearl (2000) provides the canonical treatment of directed acyclic graphs, the most common class of nonparametric graph-based causal models. Morgan and Winship (2014) includes a clear and practical exposition of the use of directed acyclic graphs for identification of causal quantities. Richardson and Robins (mimeo.) develops a generalization of directed acyclic graphs (single world intervention graphs) that provides a clear unification of graph-based and potential outcomes-based models; we recommend that readers interested in graph-based approaches to causality read it after internalizing the core ideas from Pearl (2000).